TECHNISCHE UNIVERSITÄT DARMSTADT

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Bachelorarbeit

Representational Measurement of Similarity: The Additive-Difference Model, Revisited

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Contents

Symbols

Throughout this thesis, we follow a consistent notation and try to use conventional notations and notations introduced in the used literature. Here we will provide a quick overview over the notation.

$\mathbb{R}$ and $\mathbb{Q}$ denote the real and rational numbers, respectively, with restrictions indicated via subscripts (e.g. $\mathbb{R}_{>0}$ and $\mathbb{R}_{\geq 0}$). Other sets are generally denoted with uppercase letters, except for the letters F and G, which are real-valued functions from the AD model and are used throughout this thesis.
Dimensions are represented by n and i, k and m are integers used for indexing or counting.
Vectors are bold (e.g. $\mathbf{x}, \mathbf{y}$) with coordinates $\mathbf{x} = (x_1, \ldots, x_n)$. Sequences of vectors are indexed as $\mathbf{x_i}$ or $\mathbf{x^{(i)}}$.
Greek letters are either real-valued functions and have a special meaning (such as φ, δ or γ) or constants defined in the context (e.g. α, β). Other symbols are used according to conventions, e.g. λ for convex combinations, ε for small real quantities and $|x|$ for the absolute value of x.

Any symbols not mentioned here will be introduced as needed within the relevant sections.

Metric Spaces and Topology

$\langle X, \delta \rangle$	A metric space
$\mathrm{B}^{\delta}_{(r)}$ / $\mathrm{B}^{\delta}_{[r]}$	The open(r) or closed $[r]$ ball with radius r around 0 w.r.t. the metric δ
δ	A metric function. Depending on the context, this refers to an arbitrary metric or the metric from the segmentally additive AD model
δ_r	A Minkowski (or power) metric
δ_∞	The maximum metric (Tchebychev distance)
γ	A parametric curve
$\mathrm{L}(\gamma)$	The length of a parametric curve
$\mathrm{conv}(A)$	The convex hull of A

extreme(A)	The extreme points of A

Proximity Structures

$\langle \mathcal{A}, \succsim \rangle$	A proximity structure, most of the time a factorial PS. Elements denoted by $\mathscr{a}, \mathscr{b}, \mathscr{c}$ and so on.
$\langle \mathscr{a}\mathscr{b}\mathscr{c} \rangle$	The ternary relation on a proximity structure
$\mathscr{a}\|\mathscr{b}\|\mathscr{c}$	$\mathscr{a}, \mathscr{b}$ and $\mathscr{c}$ differ only on one factor, and $\mathscr{b}$ is between $\mathscr{a}$ and $\mathscr{c}$

Functions

φ_i	A real-valued function defined on a proximity structure
F_i	The coordinate function in the AD Model

Other symbols

$\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w})$	The quasi-arithmetic ϕ-mean of $\mathbf{x}$ with weights $\mathbf{w}$
$\mathcal{M}_p(\mathbf{x}, \mathbf{w})$	For $p > 0$ the p-th power mean, for $p = 0$, the geometric mean of $\mathbf{x}$ with weights $\mathbf{w}$

Chapter 1

Introduction

1.1 Proximity Structures and Metrics

To be able to compare two stimuli pairs in a mathematical sense, we introduce a relation on a stimuli set that compares two stimuli pairs. If we have four stimuli $\mathscr{a}, \mathscr{b}, c$ and $\mathscr{d}$, then $(\mathscr{a}, \mathscr{b}) \succsim (c, \mathscr{d})$ denotes that, the stimuli $\mathscr{a}$ and $\mathscr{b}$ are less similar than the stimuli c and $\mathscr{d}$ (or more precisely, the dissimilarity between $\mathscr{a}$ and $\mathscr{b}$ is at least as great as that between c and $\mathscr{d}$). Analogously, $\sim$ means that the similarity of two pairs is equal. This essence of a *proximity structure* is exactly that, with some additional constraints.

Definition 1.1.1 (Proximity Structure)**.**
Suppose $\mathscr{A}$ is a set and $\succsim$ a quaternary relation on $\mathscr{A}$, i.e., binary relation on $\mathscr{A} \times \mathscr{A}$. $\langle \mathscr{A}, \succsim \rangle$ is a *proximity structure* (PS) iff the following axioms hold for all $\mathscr{a}, \mathscr{b} \in \mathscr{A}$:

1. $\succsim$ is a weak ordering on $\mathscr{A} \times \mathscr{A}$, i.e., it is connected and transitive.
2. $(\mathscr{a}, \mathscr{b}) \succsim (\mathscr{a}, \mathscr{a})$ whenever $\mathscr{a} \neq \mathscr{b}$.
3. $(\mathscr{a}, \mathscr{a}) \sim (\mathscr{b}, \mathscr{b})$ (minimality).
4. $(\mathscr{a}, \mathscr{b}) \sim (\mathscr{b}, \mathscr{a})$ (symmetry).

The structure is *factorial* iff

$$\mathscr{A} = \prod_{i=1}^{n} \mathscr{A}_i.$$

Remark (Connectedness)*.* Connected means, that comparison of all distinct elements is possible, i.e., for a relation R on a set X it holds for all $x, y \in X$ with $x \neq y$ that either xRy or yRx.

The mathematical concept, that we want to tie proximity structures to, is the concept of metric. A metric is defined as follows:

Definition 1.1.2 (Metric)**.**
Let X be a set. A real-valued function δ on $X \times X$ is a *metric*, iff the following properties hold for all $x, y, z \in X$:

1. $\delta(x,\ y) > 0$ unless $x = y$ and $\delta(x,\ x) = 0$
2. $\delta(x,\ y) = \delta(y,\ x)$
3. $\delta(x,\ y) + \delta(y,\ z) \geq \delta(x,\ z)$

We call $\langle X, \delta \rangle$ a *metric space.*

Metrics are measures of distance in a space. The third property is called the *triangle inequality* and is central to the intuition of measuring distances. The triangle inequality can be formulated as follows: Adding a detour through y on the way from x to z can only increase the distance.

Since we aim to connect these two concepts, we will introduce a term for that which aligns with the intuitive approach: We want the metric to preserve the structure of $\langle A, \succsim \rangle$. The only structure we have is the ordering of two stimuli pairs, and we want the metric to preserve exactly this ordering.

Definition 1.1.3 (Representability of $\succsim$ by a Metric)**.**
A proximity structure $\langle \mathscr{A}, \succsim \rangle$ is representable by a metric iff there exists a real-valued function δ on $\mathscr{A}$ such that:

1. $\langle \mathscr{A}, \delta \rangle$ is a metric space, i.e., δ is a metric on $\mathscr{A}$.

2. δ preserves the relation $\succsim$, i.e., for all $a, b, c, d \in \mathscr{A}$

$$\delta(a, b) \geq \delta(c, d) \text{ iff } (a, b) \succsim (c, d).$$

However, simple metrics do not impose enough structure, and we therefore require the metrics to satisfy additional conditions which we can observe in often used metrics. One of the most used metrics is the Minkowski p-metric or Minkowski distance (also called the power or L^p metric) and is defined in the following way.

Definition 1.1.4 (Minkowski Metric)**.**
For $n \in \mathbb{N}_{>0}$ and $p \geq 1$ the Minkowski Metric of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined as

$$\delta_p\left(\mathbf{x},\ \mathbf{y}\right) = \left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{1/p}.$$

For $p \geq 1$, this is indeed a metric, as the Minkowski inequality establishes the triangle inequality; see (Bullen, 2003, pp. 189-191 Theorem 9).

If we closely inspect this metric, we can observe two interesting properties. The first one is, comes directly from the definition of the metric. For two vectors we compute the metric by doing four steps:

1. decompose two vectors $\mathbf{x}$ and $\mathbf{y}$ into their underlying coordinates $(x_1, \ldots x_n)$ and $(y_1, \ldots, y_n)$

2. calculate the coodinate-wise differences $|x_i - y_i|$ and transform then with a bijection $F(x)$ (in this case x^p)

3. sum up the transformed differences for all coordinates

4. transform the sum using the inverse F^{-1} (in this case $x^{1/p}$)

This generalization of the metric, yields the Additive-Difference Model (Eq. (2.4)) which will be covered in Chapter 2.

The second thing we can observe in the Minkowski Metric is the fact, that distances across straight lines are additive (we will show this in Example 3.3.1). This is not the case for all metrics, but only for a special class of metrics. This class of metrics will be inspected in Chapter 3.

One interesting question is, finding conditions that assert a metric representations for a proximity structure (that preserve the $\succsim$ relation in the way we just defined). The concrete representation that are of interest to us, are the additive-difference model (which in itself is not a metric, and we therefore will combine it with a metric) and segmentally additive metrics. In both cases, this can be done by inspecting the constraints that each representation imposes on the proximity structure, and then deducing sufficient conditions for the proximity structure that allow such representations. In this thesis, we will briefly introduce and explain these conditions, but will not prove them. For that refer to (Krantz et al., 1971, Chapter 14). Instead, we will focus on the unique representation theorem for segmentally additive AD models (Theorem 4.2.1) and prove it given the representations.

Chapter 2

Additive Difference Models and Structures

In this chapter we will establish the additive difference model and additive difference structures which lead to representation with the AD model. It is based on the first four subsections of (Krantz et al., 1971, Chapter 14.4, pp. 175-185). A more elaborate explanation of the references can be found at the end of this chapter.

2.1 Additive Difference Model

As mentioned in Section 1.1 we wish to achieve a generalization of the minkowski metrics by inspecting 3 general properties that it has. Namely, these are decomposability, intradimensional subtractivity and interdimensional subtractivity which correspond to the first three steps mentioned after Definition 1.1.4 in the discussion of the minkowski metric. Here, we will define these terms for proximity structures and then remark how they translate to the reals.

Definition 2.1.1 (Additive-Difference Model).

1. *Decomposability*: The distance between stimuli is a function of componentwise contributions.

$$\delta(a, b) = F[\psi_1(a_1, b_1), \ldots, \psi_n(a_n, b_n)], \tag{2.1}$$

where F is an increasing [1] real-valued function in each of its n real arguments and ψ_i is a real-valued symmetric function satisfying $\psi_i(a_i, b_i) > \psi_i(a_i, a_i)$

[1]We use the terms increasing and decreasing to denote real-valued functions that are strictly monotonically increasing and decreasing, respectively.

when $\mathscr{a}_i \neq \mathscr{b}_i$ for all $i = 1, ..., n$.

2. *Intradimensional subtractivity*: Each componentwise contribution is the absolute value of an appropriate scale difference.

$$\delta(\mathscr{a}, \mathscr{b}) = F[|\varphi_1(\mathscr{a}_1) - \varphi_1(\mathscr{b}_1)|, \ldots, |\varphi_n(\mathscr{a}_n) - \varphi_n(\mathscr{b}_n)|], \tag{2.2}$$

   where F is as in Equation (2.1) and ψ_i is replaced with $|\varphi_i(\mathscr{a}_i) - \varphi_i(\mathscr{b}_i)|$ for some real-valued function φ_i defined on $\mathscr{A}_i$ for $i = 1, ..., n$.

3. *Interdimensional additivity*: The distance is a function of the sum of componentwise contributions.

$$\delta(\mathscr{a}, \mathscr{b}) = G\left[\sum_{i=1}^{n} \psi_i(\mathscr{a}_i, \mathscr{b}_i)\right], \tag{2.3}$$

   where ψ_i are as in Equation (2.1) and G is an increasing function in one argument.

4. *Additive-difference model*: If both, additivity and subtractivity are assumed, we get the additive-difference (AD) model.

$$\delta(\mathscr{a}, \mathscr{b}) = G\left\{\sum_{i=1}^{n} F_i\left[|\varphi_i(\mathscr{a}_i) - \varphi_i(\mathscr{b}_i)|\right]\right\}, \tag{2.4}$$

   where G and F_i, for $i = 1, \ldots, n$, are all increasing functions in one argument.

2.2 Additive Difference Structure

In this section we will develop necessary and sufficient conditions for a proximity structure to be representable by an AD Model. We do this by examining each of the characteristics of the AD model, namely decomposability, intradimensional subtractivity and interdimensional additivity, and deriving conditions for a representation. All of these conditions are closely related to the theory developed in (Krantz et al., 1989) and most of the theorems utilize some sort of reduction to the structures developed there. At the end of this subsection, we will state the representation and uniqueness theorem. As decomposability is the basis assumption of each of the variants, we begin with it.

Decomposability implies that we can represent the observed proximity of two stimuli pairs, characterized by a factorial representation, by a function that evaluates the proximity on each factors. Using the monotonicity in each argument, its necessary that each factor contributes its effect to the overall similarity independently from all the other factors. In particular, for two stimuli pairs that only differ on the i-th dimension, the ordering must be preserved, when the values on the remaining factors are changed in the same way. This is stated in the next definition, where (a, b) and (a', b') are the original stimuli and c, d, c' and d' are the stimuli we obtain by "changing" the remaining dimensions, as just described.

Definition 2.2.1 (One-Factor Independence)**.**
A factorial proximity structure $\langle A, \succsim \rangle$ satisfies one-factor independence if and only if the following holds for all $a, a', b, b', c, c', d, d' \in A$: If the two elements in each of the pairs (a, a'), (b, b'), (c, c'), (d, d') have identical components on all but one factor, and the two elements in each of the pairs (a, c), (a', c'), (b, d), (b', d') have identical components on the remaining factor, then

$$(a, b) \succsim (a', b') \quad \text{if and only if} \quad (c, d) \succsim (c', d').$$

This notion of one-factor independence induces an ordering on each factor. Therefore, one-factor independence asserts that the induced ordering is well defined, i.e., independent of the fixed components on the remaining factors. This induced ordering is denoted by $\succsim_i$. This ordering is of importance for developing intradimensional subtractivity, which we do next.

Intradimensional subtractivity is obtained by establishing decomposability for the entire structure and then imposing an absolute-difference structure from (Krantz et al., 1989, pp. 170-173, Definition 4.8) on each factor $\langle \mathcal{A}_i^2, \succsim_i \rangle$. Therefore, the

axioms for intradimensional subtractivity are essentially the same as for absolute-difference structures, but reformulated for proximity structures. In absolute-difference structures, just as in proximity structures, the basis is an ordering of pairs which is exactly $\langle \mathcal{A}_i^2, \succsim_i \rangle$ for each $i = 1, \dots, n$.
Since absolute-difference structures are unidimensional, the comparison of two pairs can be seen as the comparison of two intervals, and therefore the ordering can be seen as an "ordering of intervals". Further, the axioms for absolute-difference structures are also unidimensional. These axioms are based on the notion of unidimensional "betweenness" which is very intuitive, i.e. betweenness of real numbers, but formulated in terms of the ordering relation. The same holds for the betweenness axioms. Since our representation is in the reals, these intuitive concepts must apply to the empirical structure, and therefore are necessary conditions. Next, we present the definitions of betweenness and the betweenness axioms. Note that these can be easily verified for the reals using 6 points on a line.

Definition 2.2.2 (Betweenness and Betweenness Axioms).
Let $\langle A = \prod_{i=1}^n A_i, \succsim \rangle$ be a factorial proximity structure that satisfies one-factor independence. Let $\succsim_i$ denote the induced order on A_i^2. We say that $\mathscr{b}$ is *between* $\mathscr{a}$ and c, denoted $\mathscr{a}|\mathscr{b}|c$, if and only if

$$(\mathscr{a}_i, c_i) \succsim_i (\mathscr{a}_i, \mathscr{b}_i), (\mathscr{b}_i, c_i) \quad \text{for each } i.$$

We say that $\langle A, \succsim \rangle$ satisfies the *betweenness axioms* if and only if the following hold for all $\mathscr{a}, \mathscr{b}, c, \mathscr{d}, \mathscr{a}', \mathscr{b}', c' \in A$:

1. Suppose $\mathscr{a}, \mathscr{b}, c, \mathscr{d}$ differ on at most one factor, and $\mathscr{b} \neq c$, then

   (i) if $\mathscr{a}|\mathscr{b}|c$ and $\mathscr{b}|c|\mathscr{d}$, then $\mathscr{a}|\mathscr{b}|\mathscr{d}$ and $\mathscr{a}|c|\mathscr{d}$;

   (ii) if $\mathscr{a}|\mathscr{b}|c$ and $\mathscr{a}|c|\mathscr{d}$, then $\mathscr{a}|\mathscr{b}|\mathscr{d}$ and $\mathscr{b}|c|\mathscr{d}$.

2. Suppose $\mathscr{a}, \mathscr{b}, c, \mathscr{a}', \mathscr{b}', c'$ differ on at most one factor, $\mathscr{a}|\mathscr{b}|c$, $\mathscr{a}'|\mathscr{b}'|c'$, and $(\mathscr{b}, c) \sim (\mathscr{b}', c')$, then
$$(\mathscr{a}, c) \succsim (\mathscr{a}', c') \iff (\mathscr{a}, \mathscr{b}) \succsim (\mathscr{a}', \mathscr{b}').$$

The betweenness axioms are, however, not enough to establish a absolute-difference structure on each factor. We further need two technical axioms, which are not empirically testable. The first one is restricted solvability, which is very similar to segmental solvability but assures the existence of an element with respect to unidimensional betweenness, i.e. $a|b|c$ instead of $\langle a, b, c\rangle$. Given three stimuli

d, a and c we measure the dissimilarities with respect to d and observe the pairs (d, a) and (d, c). Then, for every possible dissimilarity between (d, a) and (d, c), there exists an element b which realizes this dissimilarity.

Definition 2.2.3 (Restricted Solvability).
A factorial proximity structure $\langle A, \succsim \rangle$ satisfies restricted solvability if and only if, for all $e, f, d, a, c \in A$, if $(d, a) \succsim (e, f) \succsim (d, c)$, then there exists $b \in A$ such that $a|b|c$ and $(d, b) \sim (e, f)$.

The second axiom that we need for intradimensional subtractivity is the archimedean axiom. The statement of this axiom can be formulated as follows: Given an distance, a radius and a center, every sequence of points starting from the center which moves away from the center with steps greater than the given distance must be finite. In other words, either the steps get smaller or the sequence moves out of the radius. It can be formulated with respect to $\succsim$ as follows.

Definition 2.2.4 (Archimedean axiom).
A proximity structure $\langle A, \succsim \rangle$ satisfies the *archimedean axiom* if and only if, for all $a, b, c, d \in A$ with $a \neq b$, any sequence $e^{(k)} \in A$, $k = 0, 1, \ldots$ such that $e^{(0)} = c$, $(e^{(k)}, e^{(k+1)}) \succ (a, b)$, and $(c, d) \succ (c, e^{(k+1)}) \succ (c, e^{(k)})$ for all k is finite.

Given these axioms, we can establish an absolute-difference structure on each factor and therefore get an intradimensionally subtractive representation.

Now, we investigate some consequences of interdimensional additivity. For two stimuli a and a' that coincide on the i-th factor, and let the same hold for b and b'. Then $\psi_i(a_i, b_i) = \psi_i(a'_i, b'_i)$, and since our representation is additive, the relation $(a, b) \succsim (a', b')$ must be independent of the i-th component. Moreover, changing the common value of a and a' and the common value of b and b' should preserve the ordering. Therefore, the ordering is independent of the common value. This independence is different from one-factor independence which we defined above and, in fact, is stronger, as it implies one-factor independence but not conversely.

Definition 2.2.5 (Independence).
A factorial proximity structure $\langle A, \succsim \rangle$ satisfies independence if the following holds for all $a, b, c, d, a', b', c', d' \in A$:

1. If the two elements inf each of the pairs $(a, a'), (b, b'), (c, c'), (d, d')$, have identical components on one factor, and
2. the two elements in each of the pairs (a, c), (b, d), (a', c'), (b', d'), have identical elements on the remaining factors, then

$$(c, d) \succsim (c', d') \iff (a, b) \succsim (a', b').$$

Given this, and that every factor is essential, i.e. each factor contributes to the ordering in the sense, that there exist $a, b \in \mathcal{A}$ that coincide on all but the one factor satisfy $(a, b) \succ (a, a)$. Using independence we can eliminate all inessential factors and assume that n denotes the number of essential factors.

Taking together all the axioms just defined yields the AD model as a representation, as the next theorem shows. Before stating the representation and uniqueness theorem, we first define additive difference structures as the structures satisfying the axioms we just derived. Note, that one factor independence is implied by independence and therefore is omitted.

Definition 2.2.6 (Additive-Difference Structure)**.**
A factorial proximity structure $\langle \mathcal{A}, \succsim \rangle$ with $n \geq 3$ Factors is an *additive-difference structure* (AD Structure), iff the following axioms are hold:

1. *Independence*
2. *Betweenness axioms*
3. *Restricted solvability*
4. *Archimedean axiom*

Theorem 2.2.7 (Representation and Uniqueness Theorem for AD Structures)**.**
Let $\langle \mathcal{A}, \succsim \rangle$ be an additive-difference factorial proximity structure. Then there exist real-valued functions $\varphi_1, \dots, \varphi_n$ defined on $\mathcal{A}_1, \dots, \mathcal{A}_n$, and increasing functions $F_1, \dots, F_n$ each defined on $\mathbb{R}$, such that for all $a, b, c, d \in \mathcal{A}$,

$$(a_1, \dots, a_n, b_1, \dots, b_n) \succsim (c_1, \dots, c_n, d_1, \dots, d_n)$$
$$\text{iff}$$
$$\sum_{i=1}^{n} F_i(|\varphi_i(a_i) - \varphi_i(b_i)|) \geq \sum_{i=1}^{n} F_i(|\varphi_i(c_i) - \varphi_i(d_i)|).$$

Furthermore, the φ_i are interval scales, and the F_i are interval scales with a common unit ($i = 1, \dots, n$).

Remark 2.2.8. For $n = 2$ another axiom, the thomsen condition, must be assumed. For $n \geq 3$ this conditions follows from the other axioms, see (Krantz et al., 1971, pp. 182-184).

2.3 Examples

Example 2.3.1 (p-exp Metric Family)**.**
Let the F in the AD model (Eq. (2.4)) be given by

$$F(x) = \alpha \cdot (e^{q \cdot x} - 1)^p \qquad \alpha, q > 0, p \geq 1. \tag{2.5}$$

The model defined by this F is a metric.

Proof. First, we will prove positivity and symmetry of this function, and then turn to the triangle inequality which is a result of theorem which provides a generalization of the Mikowski inequality.

Part 1 (Positivity and Symmetry)**.**
Positivity and symmetry are both straight forward.

1. Positivity: $\delta(\mathbf{x}, \mathbf{y}) = 0$ implies $F(|x_i - y_i|) = 0$ for all $i = 1, \ldots, n$, which is only possible if $|x_i - y_i| \cdot r = 0$, meaning $x_i = y_i$ for all $i = 1, \ldots, n$.
   Conversely, if $\mathbf{x} \neq \mathbf{y}$, then $x_i \neq y_i$ for at least one i in $1, \ldots n$, which implies $F_i(|x_i - y_i|) > 0$, and thus $\delta(\mathbf{x}, \mathbf{y}) > 0$.

2. Symmetry: Symmetry follows directly from the fact that $|x_i - y_i| = |y_i - x_i|$ holds.

The triangle inequality follows from Mullholland's Inequality (Mulholland, 1949, p. 297).

Theorem 2.3.2 (Mulholland's Inequality)**.**
The inequality

$$f^{-1}\left[\sum_{i=1}^{n} f(|x_i + y_i|)\right] \leq f^{-1}\left[\sum_{i=1}^{n} f(|x_i|)\right] + f^{-1}\left[\sum_{i=1}^{n} f(|y_i|)\right].$$

holds for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n, n = 1, 2, \ldots$ *whenever the following three conditions are satisfied:*

1. f *is continuous, increasing and* $f(0) = 0$.

2. f *is convex on* $[0, \infty)$.

3. $\log f(e^x)$ *is convex for* $x \in (-\infty, \infty)$.

Part 1 (Proof that Eq. (2.5) satisfies Mulholland's conditions)**.**
Regarding Point 1: Continuity and $f(0) = 0$ are clear. For monotonicity, note that $e^{qx} - 1$ is increasing, exponentiation is also increasing, and multiplying by $\alpha > 0$ preserves monotonicity. Thus, the composition is also increasing.
Regarding point 2: Similar reasoning applies for convexity. The composition of convex functions is convex, if the outer function is non-decreasing (Thomson et al., 2008, p. 481, Note 207). Since multiplication by a positive constant and exponentiation for exponents greater than 1 is both convex and increasing, the resulting function must be convex.

We now proceed to prove the convexity of $\log F(e^x)$. By logarithmic properties, it holds that

$$\log \alpha(e^{qe^x} - 1)^p = \log \alpha + p \cdot \log(e^{qe^x} - 1).$$

Arbitrary addition and multiplication by $p \geq 1$ preserve convexity, so it suffices to show the convexity of $\log(e^{qe^x} - 1)$. We prove this by examining the second derivative. If the second derivative is strictly positive, the function is convex (Hug & Weil, 2020, p. 57, Remark 2.13). Differentiating twice yields

$$\frac{qe^{qe^x+x}\left(q(-e^x) + e^{qe^x} - 1\right)}{\left(e^{qe^x} - 1\right)^2} \overset{!}{>} 0.$$

The denominator is always positive and non-zero, so we can ignore it. Similarly, the factor qe^{qe^x+x} is always positive and can also be omitted. It remains to show that

$$-qe^x + e^{qe^x} - 1 > 0$$
$$\Updownarrow$$
$$e^{qe^x} - qe^x > 1$$

holds. This can be shown in two steps. First, we show that the left-hand side is strictly increasing. Second, we prove that its limit as $x \to -\infty$ is 1. Therefore, the inequality holds for all $x \in \mathbb{R}$.

1. For the first point, we show that the derivative is greater than zero. Differentiation yields
$$qe^x \cdot e^{qe^x} - qe^x = qe^x \cdot (e^{qe^x} - 1) > 0.$$
The inequality holds, since $e^x > 0$ for all $x \in \mathbb{R}$ and $e^x > 1$ for $x > 0$.

2. For the second point, we calculate the limit as $x \to -\infty$:
$$\lim_{x\to-\infty} e^{qe^x} - qe^x = \lim_{x\to-\infty} e^{qe^x} - \lim_{x\to-\infty} qe^x = 1 - 0 = 1.$$

Thus, convexity is established, and the claim follows from Mulholland's theorem.

□

Remark 2.3.3 (Minkowski metric is part of the p-exp metric family). If we set $\alpha = q^{-p}$ in (Eq. (2.5)), we obtain the Minkowski metric in the limit as $q \to 0$.

Proof. Setting $\alpha = q^{-p}$ yields

$$q^{-p}(e^{qx} - 1)^p = \left(\frac{e^{qx} - 1}{q}\right)^p.$$

As $q \to 0$, both the numerator and denominator tend to 0 (we can pull the limit inside the exponent since the exponentiation is continuous), therefore we can apply L'Hôpital's rule. Hence

$$\begin{aligned}
\lim_{q\to 0}\left(\frac{e^{qx} - 1}{q}\right)^p &= \left(\lim_{q\to 0}\frac{e^{qx} - 1}{q}\right)^p && \text{(continuity)}\\
&= \left(\lim_{q\to 0}\frac{x \cdot e^{qx} - 1}{1}\right)^p && \text{(L'Hôpital's rule)}\\
&= x^p.
\end{aligned}$$

□

Not every function that can be represented by the additive difference model is a metric.

Sources and Original Contributions

A more elaborate analysis can be found in (Krantz et al., 1971, Chapter 14.4, pp. 175-185), where for each of the variants of the AD model a representation and uniqueness theorem is provided. Here we only focus on the AD Model, which is the combination of decomposability, interdimensional additivity and intradimensional subtractivity.

Decpomosability and $\succsim_i$ is closely related to the Theory of conjoint measurement, specifically decomposable conjoint measurement, developed in (Krantz et al., 1989, Chapter 7).

Chapter 3

Segmental Additivity

3.1 Segmentally Additive Metrics

The definitions and statements from this section are taken out from (Busemann, 1955, Chapter 1).

Definition 3.1.1 (Curves and Segments)**.**

1. Let δ be a metric on X. A parametric curve in $\langle X, \delta \rangle$ is a function γ that maps a closed interval $[a, b] \subset \mathbb{R}$ into X and is continuous. The parametric curve γ is said to join $\gamma(a)$ and $\gamma(b)$.

2. Let γ be a parametric curve in $\langle X, \delta \rangle$ with domain $[a, b]$. Its length $\mathrm{L}(\gamma)$ is the supremum of the sums

$$\sum_{i=1}^{n} \delta \left[\gamma(a_{i-1}), \gamma(a_i)\right]$$

   over all finite sequences $a_0, a_1, \ldots, a_n$ such that

$$a = a_0 \leq a_1 \leq \cdots \leq a_n = b.$$

   If $\mathrm{L}(\gamma) < \infty$, then γ is rectifiable.

3. A rectifiable curve is a segment iff its length is equal to the distance between its endpoints, i.e. $\mathrm{L}(\gamma) = \delta\,(a,\ b)$.

4. For $a \leq x \leq b$ we denote the length of the sub-curve $\gamma(t),\ a \leq t \leq x$ from a to x by $\mathrm{L}_a^x(\gamma)$. This is also called the *arc length* of the curve.

Remark 3.1.2 (Continuity & Monotonicity of Sub Length). For a rectifiable curve $\gamma(t),\ a \leq t \leq b$, the arc length $\mathrm{L}_a^x(\gamma)$ is a continuous, non-decreasing function of x.

Definition 3.1.3 (Metric Space with Additive Segments)**.**
Let δ be a metric on X. $\langle X, \delta\rangle$ is a metric space with additive segments iff for any $x, y \in X$ there is a segment joining them.

3.2 Segmentally Additive Proximity Structures

Definition 3.2.1 (Ternary Relation on Proximity Structures).
Let $\langle \mathcal{A}, \succsim \rangle$ be a factorial proximity structure. For $a, b, c \in \mathcal{A}$, the ternary relation $(a, b, c\rangle$ is said to hold iff for all $a', b', c' \in \mathcal{A}$:

1. If $(a, b) \succsim (a', b')$ and $(a', c') \succsim (a, c)$, then $(b', c') \succsim (b, c)$
2. If either of the preceding inequalities is strict, then the resulting inequality is also strict.

Further the relation $\langle abc\rangle$ is said to hold if both $(a, b, c\rangle$ and $(c, b, a\rangle$ hold.

Definition 3.2.2 (Segmentally Additive Proximity Structure).
A proximity structure $\langle \mathcal{A}, \succsim \rangle$, with a ternary relation $\langle\rangle$ defined in terms of $\succsim$, is segmentally additive if and only if the following two axioms hold for all $a, c, e, f \in \mathcal{A}$:

1. If $(a, c) \succsim (e, f)$, then there exists $b \in \mathcal{A}$ such that $(a, b) \sim (e, f)$ and $\langle abc\rangle$. (segmental solvability)
2. If $e \neq f$, then there exist $b_0, \ldots, b_n \in \mathcal{A}$, such that $b_0 = a$, $b_n = c$, and $(e, f) \succsim (b_{i-1}, b_i)$ for $i = 1, \ldots, n$. (connectedness)

A segmentally additive proximity structure is complete if and only if the following two axioms hold for all $a, c, e, f \in \mathcal{A}$:

3. If $e \neq f$, then there exist $b, d \in \mathcal{A}$, with $b \neq d$, such that $(e, f) \succ (b, d)$. (Non-discreteness of distances)
4. Let $a_1, \ldots, a_n, \ldots$ be a sequence of elements of $\mathcal{A}$, such that for all $e, f \in \mathcal{A}$, if $e \neq f$, then $(e, f) \succsim (a_i, a_j)$ for all but finitely many pairs (i, j). Then there exists $b \in \mathcal{A}$, such that for all $e, f \in \mathcal{A}$, if $e \neq f$, then $(e, f) \succsim (a_i, b)$ for all but finitely many i. (Limits for Cauchy sequences)

Theorem 3.2.3 (Representation and Uniqueness Theorem for Segmentally Additive Proximity Structures).
Suppose $\langle \mathcal{A}, \succsim \rangle$ is a segmentally additive proximity structure. Then $\langle \mathcal{A}, \succsim \rangle$ is representable by a metric, and additionally, for all $a, b, c \in \mathcal{A}$:

1. *$\langle abc\rangle$ iff $\delta(a, b) + \delta(b, c) = \delta(a, c)$.*
2. *If δ' is another metric on $\mathcal{A}$ that satisfies the above conditions, then there exists $\alpha > 0$ such that $\delta' = \alpha\,\delta$ (δ is a ratio scale).*

3.3 Examples

Example 3.3.1 (Segmental Additivity of the Minkowski Metric)**.**
For $p \geq 1$ the Minkowski Metric is a metric with additive segments. The segments are all straight lines (i.e. the line segments between two arbitrary points).

Proof. Let $\mathbf{x}, \mathbf{z} \in \mathbb{R}^n$ and $0 \leq t \leq 1$. Then, for $\mathbf{y} = (1-t)\mathbf{x} + t\mathbf{z}$ ($\mathbf{y}$ is on the line segment between $\mathbf{z}$ and $\mathbf{x}$, iff $\mathbf{y}$ can be represented in this way) it holds that:

$$
\begin{aligned}
\delta_p(\mathbf{x},\ \mathbf{y}) + \delta_p(\mathbf{y},\ \mathbf{z}) &= \left[\sum_{i=1}^{n} |x_i - (1-t)x_i - tz_i|^p\right]^{1/p} + \left[\sum_{i=1}^{n} |(1-t)x_i + tz_i - z_i|^p\right]^{1/p} \\
&= \left[\sum_{i=1}^{n} |x_i - x_i + tx_i - tz_i|^p\right]^{1/p} + \left[\sum_{i=1}^{n} |x_i - tx_i + tz_i - z_i|^p\right]^{1/p} \\
&= t\left[\sum_{i=1}^{n} |x_i - z_i|^p\right]^{1/p} + (1-t)\left[\sum_{i=1}^{n} |x_i - z_i|^p\right]^{1/p} \\
&= \delta_p(\mathbf{x},\ \mathbf{z}).
\end{aligned}
$$

□

Sources and Original Contributions

Test test 4

Chapter 4

Combining Segmental Additivity with the AD Model

4.1 Conditions for Metric Representation of AD Structures

Theorem 4.1.1 (Conditions for Metric Representation of AD Structures).
Suppose $\langle \mathcal{A}, \succsim \rangle$ is an additive-difference proximity structure such that f is the additive-difference representation from Theorem 2.2.7, i.e.,

$$f(a, b) = \sum_{i=1}^{n} F_i(|\varphi_i(a) - \varphi_i(b)|).$$

Suppose further that the domain of each F_i is a real interval and $F_i(0) = 0$, $i = 1, \ldots, n$ (through transformation, since F_i are interval scales).

For $\langle \mathcal{A}, \succsim \rangle$ to be compatible with a metric, and additionally be additive on the coordinate axes, it is

1. *necessary that there exists a continuous, convex function F, such that $F_i(x) = F(t_i x), t_i > 0$, for all x in the domain of F_i, $i = 1, \ldots, n$; and*
2. *sufficient that, in addition (to the conditions in 1.), the function $\log F(e^x)$ is convex.*

Remark 4.1.2 (Additivity on the Coordinate Axes). Additivity on the coordinate axes means the following: If $a, b, c \in \mathcal{A}$ differ only on one factor $\mathcal{A}_i$, then the triangle inequality of the metric holds with equality. In mathematical terms:

$$a_j = b_j = c_j \text{ for all } j \neq i \text{ and } (a, c) \succsim (a, b), (b, c)$$

implies that $\delta(a,\ b) + \delta(b,\ c) = \delta(a,\ c)$. This can be also formulated in terms of betweenness (Definition 2.2.2). If a, b and c differ only on one factor and $a|b|c$, then $\delta(a,\ b) + \delta(b,\ c) = \delta(a,\ c)$

In the following lemmas, until the proof of this theorem is finished, everything will be assumed as in Theorem 4.1.1, unless otherwise stated. The following remark ties this result closer to the Minkowski metric, and provides a form which the metric must take. This will play an important role in the main result Theorem 4.2.1.

Remark 4.1.3 (Form of δ). Examining Section 4.1 more closely yields, that $G = F^{-1}$ must hold and the representation of δ takes the form

$$\delta(a, b) = F^{-1}\Big\{\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\}$$

where F is continuous, convex and increasing. This follows form two arguments.

Firstly, incorporating the t_i into the φ_i (which is possible because, by Theorem 2.2.7, the φ_i are interval scales)

$$\begin{aligned} G\Big\{\sum_{i=1}^{n} F_i(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} &= G\Big\{\sum_{i=1}^{n} F(t_i|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} \\ &= G\Big\{\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|)\Big\} \end{aligned}$$

yields the desired form inside the sum.

Secondly, by using $F_i(x) = F_n(t_i x)$ and $F_i(x) = G^{-1}(t_i x)$. The first equation applied for $i = n$ yields $F_n(x) = F_n(t_n x)$ which implies $t_n = 1$. Using this in the second equation for $i = n$ yields $F(x) = F_n(x) = G^{-1}(t_n x) = G^{-1}(x)$. Hence, by taking the inverse on both sides (which is possible due to F and G both being increasing), we get the desired form for δ.

Proof of Theorem 4.1.1 1. According to the assumptions of the theorem, the additive difference representation $f(a, b)$ is compatible with a metric δ on $\mathcal{A}$, i.e.,

$$f(a, b) \leq f(c, d) \quad \text{iff} \quad \delta(a, b) \leq \delta(c, d) \quad \text{for all } a, b, c, d \in \mathcal{A}.$$

Therefore, there exists an increasing function G [1], such that for all $a, b \in \mathcal{A}$

$$\delta(a, b) = G(f(a, b)) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right).$$

[1] A possible perspective on this is, that the metric δ induces an ordinal scale for pairs of elements. The same holds for the function f and these orderings are the same. Because ordinal scales, by definition, are invariant under increasing bijections, there must exist an increasing function G, which transforms the f-scale into the δ-scale.

Assume a, b, and c differ only in the i-th factor and $a|b|c$. Thus, we can use additivity on the i-th coordinate axis for $a, b, c \in \mathcal{A}$ which yields

$$G\left(F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) + G\left(F_i\left(|\varphi_i(b_i) - \varphi_i(c_i)|\right)\right) = G\left(F_i\left(|\varphi_i(a_i) - \varphi_i(c_i)|\right)\right).$$

Defining $|\varphi_i(a_i) - \varphi_i(b_i)| =: x$ and $|\varphi_i(b_i) - \varphi_i(c_i)| =: y$ yields

$$G[F_i(x)] + G[F_i(y)] = G[F_i(x+y)].$$

Further, define $H_i(x) = G[F_i(x)]$. This reduces the above equation to $H_i(x) + H_i(y) = H_i(x+y)$ for all x, y in the domain of $F_i, 1 \leq i \leq n$. By Theorem 2.2.7, the domain is $\mathbb{R}$. This is Cauchy's functional equation and, under the assumption of monotonicity, the only solution is $H_i = t_i x$ for $t_i \in \mathbb{R}$, as shown in (Aczel & Dhombres, 1989, p. 15). Since the values are positive (as we are dealing with metrics), $H_i = t_i x$ with $t_i > 0$ must hold for all $i = 1, ..., n$. Additionally, since G is increasing, it has an inverse G^{-1} satisfying

$$G^{-1}[H_i(x)] = G^{-1}[t_i x] = F_i(x)$$

where F_i is defined. Since G^{-1} is independent of i, the F_i can only differ in range and in the unit of the domain (i.e. the scaling factor t_i which defines the domain of F_i by adjusting the domain of G^{-1}). Without loss of generality, we can assume that the range of F_n is the largest among the F_i. This assumption allows us to define $F = F_n$ as the reference function, and the other F_i can be expressed as $F_i(x) = F_n(t_i x) = F(t_i x)$ for $t_i > 0$ and for every x in the domain of F_i. Continuity and convexity will be proven in the next lemmas. □

Lemma 4.1.4.

$$F(x+y+z) + F(z) \geq F(x+z) + F(y+z).$$

Proof. In the triangle inequality,

$$F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] + F^{-1}\left[\sum_{i=1}^{n} F(y_i)\right] \geq F^{-1}\left[\sum_{i=1}^{n} F(x_i + y_i)\right],$$

let $x_1 = z, x_2 = F^{-1}(F(x+z) - F(z)), x_j = 0$ for $3 \leq j \leq n$ and $y_1 = y, y_j = 0$ for $2 \leq j \leq n$. Hence,

$$\begin{aligned} F(x+y+z) &= F\left[F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] + F^{-1}\left[\sum_{i=1}^{n} F(y_i)\right]\right] \\ &\geq \sum_{i=1}^{n} F(x_i + y_i) \qquad \text{(by the triangle inequality)} \\ &= F(y+z) + F(x+z) - F(z). \end{aligned}$$

□

Lemma 4.1.5.
F is continuous.

Proof. Since F is increasing and defined on an interval, all discontinuities are gaps (Thomson et al., 2008, p. 335) [2]. Real analysis yields, that real monotonic functions have at most countable many gaps. We will derive a contradiction from this. Assume $F(x)$ is the lower boundary of a gap, then there exists $\varepsilon > 0$ such that for all $\delta > 0$ [3], it must hold that $F(x + \delta) - F(x) > \varepsilon$. However, by Lemma 4.1.4, for every $y > 0$,

$$F(x + y + \delta) - F(x + y) \geq F(x + \delta) - F(x) > \varepsilon,$$

and so there is a gap at any $x+y$, and F would have to be discontinuous everywhere. This is impossible, so F must be continuous. □

Lemma 4.1.6 (Characterization of Convexity)**.**
Suppose F is a strictly increasing function defined on the nonnegative reals. Then F is convex iff for all $x, y, z \geq 0$ the inequality from Lemma 4.1.4 holds, i.e.,

$$F(x + y + z) + F(z) \geq F(x + z) + F(y + z).$$

Proof.

Part 1 ($\Rightarrow$)**.**
Suppose, F is convex, and let $\lambda = x/(x + y)$, then

$$\begin{aligned} \lambda(x + y + z) + (1 - \lambda)z &= x + z \\ (1 - \lambda)(x + y + z) + \lambda z &= y + z. \end{aligned}$$

Hence, by convexity,

$$\begin{aligned} F(x + z) + F(y + z) &= F\left[\lambda(x + y + z) + (1 - \lambda)z\right] + F\left[(1 - \lambda)(x + y + z) + \lambda z\right] \\ &\leq \lambda F(x + y + z) + (1 - \lambda)F(z) + (1 - \lambda)F(x + y + z) + \lambda F(z) \\ &= F(x + y + z) + F(z). \end{aligned}$$

---

[2](Thomson et al., 2008) prove the following statement: Let f be a real-valued monotonic function defined on an interval $[a, b]$. Then the set of points of discontinuity of f in that interval is countable. Combining this statement with the paragraph above it, yields that these discontinuities must be gaps. Since any interval (open, half-open and unbouded intervals) can be represented by a countable union of closed intervals, this theorem can be extended to arbitrary intervals since countable unions of countable sets are still countable. This can be done by using $(a, b) = \bigcup_{n=1}^{\infty}[a + 1/n, b - 1/n]$, or $(a, \infty) = \bigcup_{n=1}^{\infty}[a + 1/n, a + n]$. Similar arguments apply to other types of intervals.

[3]Here, δ differs from our usual usage and is not a metric; rather it follows the standard notation for continuity and is only used in this proof this way.

Part 2 ($\Leftarrow$).

To prove the converse, we will use the continuity of F by Lemma 4.1.5. Suppose, $x > y$, then

$$\begin{aligned} F(x) + F(y) &= F\left(\frac{x-y}{2} + \frac{x-y}{2} + y\right) + F(y) \\ &\geq 2F\left(\frac{x-y}{2} + y\right) && \text{(by Lemma 4.1.5)} \\ &= 2F\left(\frac{x+y}{2}\right), \end{aligned}$$

which implies convexity for a continuous F (Hug & Weil, 2020, p. 51, Exercise 6).

$\square$

4.2 Unique Representation Theorem

Theorem 4.2.1 (Unique Representation Theorem: The Power Metric as the Sole Representation for Segmentally Additive AD Structures)**.**
Suppose $\langle \mathcal{A}, \succsim \rangle$ is a complete, segmentally additive, factorial proximity structure, that is also and additive difference structure. Then there exists a unique $p \geq 1$ and real-valued functions φ_i, defined on $\mathcal{A}_i$, $i = 1, \ldots, n$, that are unique up to a positive linear transformation, such that, for all $a, b, c, d \in \mathcal{A}$,

$$(a, b) \succsim (c, d) \quad \textit{iff} \quad \delta(a, b) \geq \delta(c, d),$$

where

$$\delta(a, b) = \left[\sum_{i=1}^{n} |\varphi_i(a) - \varphi_i(b)|^p \right]^{1/p}.$$

Proof. This proof proceeds in multiple steps. First, we make observations about the domains and ranges of the functions F_i, use this to find bounds for the domain of F. Using these bounds, we argue that F can be extended to the positive reals. Next, we derive from this a functional equation, that is only satisfied by $f(x) = \alpha x^p$ for $\alpha > 0$. Finally, we prove that F is equal to this extension on the whole domain, and therefore δ is a Minkowski metric.

Part 0 (Establishing the basics and using the Theorem 4.2.3)**.**
By Theorem 4.1.1 and Remark 4.1.3, δ must be of the form

$$\delta\left(a,\ b\right) = F^{-1} \left[\sum_{i=1}^{n} F(|\varphi_i(a_i) - \varphi_i(b_i)|) \right]$$

where F is continuous, convex and increasing. By the same argument as in Remark 4.2.4 (iii), δ must be defined on a cartesian product of intervals of the form $[0, \rho_i)$ where ρ_i are the ranges of $|\varphi_i(a_i) - \varphi_i(b_i)|$ for $i = 1, \ldots, n$. This cartesian product is convex and, by leaving out 0 in each product, also open. Therefore the requirements for Theorem 4.2.3 are met if we leave out $\mathbf{0}$ and applying the lemma yields homogeneity for for $\mathbf{x}, \mathbf{z} \neq \mathbf{0}$:

$$\delta\left(\mathbf{x},\ \mathbf{x} + t(\mathbf{z} - \mathbf{x})\right) = t\,\delta\left(\mathbf{x},\ \mathbf{z}\right).$$

Applying homogeneity for the definition of δ above and setting $x_i = |\varphi_i(b_i) - \varphi_i(a_i)|$ yields

$$F^{-1} \left[\sum_{i=1}^{n} F(t x_i) \right] = t \cdot F^{-1} \left[\sum_{i=1}^{n} F(x_i) \right]. \tag{4.1}$$

However, due to translation invariance, homogeneity also holds for $\mathbf{x} = \mathbf{0}$ [4]. We will show, that the only solutions to this equation are of the form $F(x) = \alpha x^p$, i.e. the power functions.

First, we start by observing the domains and the ranges of F. By Theorem 4.1.1, the additive difference model

$$\delta(a, b) = G\left(\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right)$$

yields a metric (that is additive on the coordinate axes) if $F_i(x) = F(t_i x)$ and $F = G^{-1}$ for $i = 1, ..., n$. Thus, F^{-1} is defined for all numbers $\sum_{i=1}^{n} F_i\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)$, and each F_i coincides with F on its domain (except for the factor t_i, which scales the argument and adjusts the domain). Since the domains of F_i are intervals (because δ is a metric with additive segments and therefore for any point in the domain of F_i, the interval between zero and that point must also be in the domain), we can choose $\omega > 0$ such that for each $i = 1, \ldots, n$ and any x with $0 \le x \le \omega/t_i$, x lies within the domain of F_i. In particular, if we set $\omega_i = \omega/t_i$, then $F^{-1}\left[\sum_{i=1}^{n} F_i(\omega_i)\right] = F^{-1}\left[nF(\omega)\right] =: \Omega$ is defined. Thus, $F = G^{-1}$ is defined at least on the interval $[0, \Omega]$. We can now extend F as follows.

Part 1 (Extending the bounds of F)**.**
For $x < \omega$, we have by Equation (4.1),

$$\begin{aligned} F^{-1}[nF(x)] &= F^{-1}[nF(x\omega/\omega)] \\ &= (x/\omega)F^{-1}[nF(\omega)] \\ &= x\Omega/\omega. \end{aligned}$$

Thus F satisfies the following, by plugging in the correct values for x ($x\frac{\omega}{\Omega}$ and $F^{-1}(\frac{x}{n})$)

$$F(x) = nF(x\omega/\Omega), \quad 0 \le x \le \Omega; \tag{4.2}$$
$$F^{-1}(x) = (\Omega/\omega)F^{-1}(x/n), \quad 0 \le x \le nF(\omega). \tag{4.3}$$

These two equations can now be used to extend the domain of F and F^{-1}.

We use Equation (4.2) by defining F through the right-hand side (for values where the right-hand side is defined, i.e., for $0 \le x \le \Omega^2/\omega$). Since $\Omega^2/\omega > \Omega$, this

[4] By adding $0 = \mathbf{y} - \mathbf{y}$ for some $\mathbf{y}$ in the domain, we get homogeneity for $\mathbf{0}$:

$$\delta(\mathbf{0},\ t\mathbf{z}) = F^{-1}\left[\sum_{i=1}^{n} F(t(y_i - (y_i - z_i)))\right] = tF^{-1}\left[\sum_{i=1}^{n} F(y_i - (y_i - z_i))\right] = t\,\delta(\mathbf{0},\ \mathbf{z}).$$

indeed extends F beyond the interval $[0, \Omega]$. Similarly, F^{-1} satisfies Equation (4.3) for the extended interval $0 \leq x \leq nF(\Omega) = n^2F(\omega)$. Moreover, Equation (4.1) is now valid for $0 \leq t \leq 1$ and $0 \leq x_i \leq \Omega$:

$$\begin{aligned}
F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} nF(tx_i\omega/\Omega)\right] && \text{(by Eq. (4.2))} \\
&= F^{-1}\left[n\sum_{i=1}^{n} F(tx_i\omega/\Omega)\right] \\
&= \frac{\Omega}{\omega}F^{-1}\left[\sum_{i=1}^{n} F(tx_i\omega/\Omega)\right] && \text{(by Eq. (4.3))} \\
&= \frac{\Omega}{\omega}tF^{-1}\left[\sum_{i=1}^{n} F(x_i\omega/\Omega)\right] \\
& && \text{(by homogeneity applied for } x_i\omega/\Omega \leq \omega) \\
&= tF^{-1}\left[n\sum_{i=1}^{n} F(x_i\omega/\Omega)\right] && \text{(by Eq. (4.3))} \\
&= tF^{-1}\left[\sum_{i=1}^{n} nF(x_i\omega/\Omega)\right] \\
&= tF^{-1}\left[\sum_{i=1}^{n} F(x_i)\right]. && \text{(by Eq. (4.2))}
\end{aligned}$$

Further, the extended function is both increasing and continuous since the extension is based on scaling of the domain and the range of the original function.

For continuity let $x_k \to x$ with $\omega < x < \Omega$. Then

$$\begin{aligned}
\lim_{k\to\infty} F(x_k) &= \lim_{k\to\infty} nF\left(x_k\frac{\omega}{\Omega}\right) && \text{(Definition of } F(t) \text{ for } \omega \leq t \leq \Omega) \\
&= nF\left(\lim_{k\to\infty} x_k\frac{\omega}{\Omega}\right) && \text{(Continuity of } F(t) \text{ for } 0 \leq t \leq \omega) \\
&= nF\left(x\frac{\omega}{\Omega}\right) && \text{(Convergence of } x_k) \\
&= F(x) && \text{(Definition of } F(t) \text{ for } \omega \leq t \leq \Omega)
\end{aligned}$$

which is equivalent to F begin continuous on the extended domain.

For monotonicity we can use a similar argument. Let $x < x'$ with $\omega < x, x' < \Omega$. Then $x\frac{\omega}{\Omega} < x'\frac{\omega}{\Omega}$ holds. Plugging that into F and multiplying by n yields $nF\left(x\frac{\omega}{\Omega}\right) < nF\left(x'\frac{\omega}{\Omega}\right)$ due to F being increasing on the original domain. Therefore $x < x'$ implies $F(x) < F(x')$ which is equivalent to F increasing. Thus, we have established strict increasing monotonicity, continuity and Equation (4.1) for the extended function.

Repeatedly applying the arguments, extends the intervals in which F is defined and in which monotonicity, continuity and homogeneity (Eq. (4.1)) are valid by a factor of Ω/ω each time. Thus, we can extend F to $[0,\infty)$ while preserving these properties.

We can prove further, that the homogeneity equation must also hold for $t \in [0,\infty)$. For that, let $x_i > 0$ and $t > 1$. Then there exists $t' = \frac{1}{t} < 1$ such that $t' \cdot tx_i = x_i$. Therefore

$$\begin{aligned} tt'F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] && (tt' = 1) \\ &= F^{-1}\left[\sum_{i=1}^{n} F(tt'x_i)\right] && (tt' = 1) \\ &= t'F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right] \end{aligned}$$

The last step holds, because both $t' < 1$ and $tx_i < \infty$ hold. Dividing both sides by t' yields

$$F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] \qquad \text{for } x, t < \infty.$$

To sum up, we have obtained an extended function which satisfies monotonicity, continuity and homogeneity for $t \in [0,\infty)$. This extended function must coincide with tie original F on at least the interval $[0,\Omega]$ (because this was the domain of F that we deduced, before extending it).

Part 2 (Deriving functional equation)**.**
Next, we derive a functional equation that allows us to use results on quasi-arithmetic homogeneous means (Hardy et al., 1952, p. 68). The theorem states as follows:

Theorem (Uniqueness of homogeneous quasi-arithmetic means, Theorem 4.2.19)**.** *Suppose that ϕ is continuous and strictly increasing in the open interval $(0,\infty)$ and that*

$$\mathcal{M}_\phi(t\mathbf{x},\ \mathbf{w}) = t\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w}) \quad \textit{with} \quad \mathcal{M}_\phi(\mathbf{x},\ \mathbf{w}) = \phi^{-1}\left(\sum_{i=1}^{n} w_i\phi(x_i)\right)$$

for all positive $\mathbf{x}, \mathbf{w}, t$ and all $n \in \mathbb{N}$ such that $\sum_{i=1}^{n} w_i = 1$. Then $\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w}) = \mathcal{M}_p(\mathbf{x}, \mathbf{w})$. (i.e. $\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w})$ is either a power mean with $\phi(x) = \alpha x^p + \beta$ and $\alpha, \beta, p \in \mathbb{R}, p > 0$ or a geometric mean with $\phi(x) = \alpha \log(x)$, $\alpha \in \mathbb{R}$.)

The proof of this statement will be presented in Section 4.2.2, here we will apply the theorem for our proof.

Since all requirements for ϕ are already met by F, we have to derive the functional equation for all $n \in \mathbb{N}$ and all positive weights. First, we will prove that the homogeneity holds not only for sums with n summands, but for arbitrary many summands $N \in \mathbb{N}$. Then we will incorporate equal weights into this equation, extend them to commensurable weights (i.e. rational weights with a common denominator) and then use this to approximate arbitrary real weights. blablabla thist

Subpart 2.1 (Homogeneity for arbitrary long sums)**.**
Let for $N < n$ the first N coordinates of $\mathbf{x}$ be zero, i.e. $\mathbf{x} = (x_1, \dots, x_N, 0, \dots, 0)$. Then, since $F(0) = 0$ it holds that, $F(x_i) = 0$ for $i > N$. Applying homogeneity for this $\mathbf{x}$ yields

$$\begin{aligned} F^{-1}\left[\sum_{i=1}^{N} F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{n} F(tx_i)\right] && (x_i = 0 \text{ for } i > N) \\ &= tF^{-1}\left[\sum_{i=1}^{n} F(x_i)\right] && \text{(homogeneity)} \\ &= tF^{-1}\left[\sum_{i=1}^{N} F(x_i)\right]. && (x_i = 0 \text{ for } i > N) \end{aligned}$$

Therefore, we have homogeneity for $N \leq n$:

$$F^{-1}\left[\sum_{i=1}^{N} F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{N} F(x_i)\right]. \tag{4.4}$$

Now, we proceed with homogeneity for $n+1$ summands. This can be achieved by grouping the last two summands into one and applying homogeneity twice:

$$\begin{aligned} F^{-1}\left[\sum_{i=1}^{n+1} F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{n-1} F(tx_i) + F(tx_n) + F(tx_{n+1})\right] && \text{(expanding sum)} \\ &= F^{-1}\left[\sum_{i=1}^{n-1} F(tx_i) + F(F^{-1}(F(tx_n) + F(tx_{n+1})))\right] \\ & && \text{(regrouping)} \\ &= F^{-1}\left[\sum_{i=1}^{n-1} F(tx_i) + F(tF^{-1}(F(x_n) + F(x_{n+1})))\right] \\ & && \text{(Eq. (4.4) on the last summand, } N = 2) \\ &= tF^{-1}\left[\sum_{i=1}^{n-1} F(x_i) + F(F^{-1}(F(x_n) + F(x_{n+1})))\right] \\ & && \text{(homogeneity for } n \text{ summands)} \\ &= tF^{-1}\left[\sum_{i=1}^{n-1} F(x_i) + F(x_n) + F(x_{n+1})\right] && \text{(cancellation of inverses)} \end{aligned}$$

$$= tF^{-1}\left[\sum_{i=1}^{n+1} F(x_i)\right].$$

Therefore, we now have homogeneity for $n+1$ summands. Repeating this, yields homogeneity for arbitrary many summands:

$$F^{-1}\left[\sum_{i=1}^{N} F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{N} F(x_i)\right] \quad N \in \mathbb{N}. \tag{4.5}$$

Subpart 2.2 (Homogeneity with weights)**.**
Now, we proceed with incorporating weights into the equation. Firstly, applying Equation (4.3) on the left hand side of Equation (4.5) yields

$$\Omega/\omega F^{-1}\left[\frac{1}{n}\sum_{i=1}^{N} F(tx_i)\right] = F^{-1}\left[\sum_{i=1}^{N} F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{N} F(x_i)\right].$$

Applying Equation (4.3) once more on the right hand side, yields

$$\Omega/\omega F^{-1}\left[\frac{1}{n}\sum_{i=1}^{N} F(tx_i)\right] = F^{-1}\left[\sum_{i=1}^{N} F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{N} F(x_i)\right] = \Omega/\omega tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{N} F(x_i)\right].$$

Hence

$$F^{-1}\left[\frac{1}{n}\sum_{i=1}^{N} F(tx_i)\right] = tF^{-1}\left[\frac{1}{n}\sum_{i=1}^{N} F(x_i)\right]$$

must hold for all $N \in \mathbb{N}$. Repeatedly applying this argument yields that

$$F^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{N} F(tx_i)\right] = tF^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{N} F(x_i)\right] \tag{4.6}$$

must hold for all $l, N \in \mathbb{N}$.

Now we extend this to weights of the form $\frac{q}{n^l}$ with $q, l \in \mathbb{N}$. For this, denote the set of all these numbers as $\mathbb{W} := \{\frac{q}{n^l} : q, l \in \mathbb{N}\}$. Let $\mathbf{w} \in \mathbb{W}^N$. Then we can write $w_i = \frac{q_i}{n^{l_i}}$ for all $i = 1, \ldots, N$. We can assume that l_i is the same for all i without loss of generality, by having n_l be a common denominator and expanding all fractions accordingly. Therefore, we not have weights $w_i = \frac{q_i}{n^l}$ for $i = 1, \ldots N$. The basic idea now is, to expand each summand $q_i \cdot F(tx_i)$ into a sum of $F(tx_i)$ with length q_i. For that, let $Q = \sum_{i=1}^{N} q_i$ and define $\tilde{x}_i$ by repeating x_i for q_i times:

$$\tilde{x}_i = \begin{cases} x_1 & i = 1, \ldots, q_1 \\ x_2 & i = q_1 + 1, \ldots, q_2 \\ & \vdots \\ x_N & i = q_{N-1} + 1, \ldots, q_N \end{cases}.$$

Then

$$\begin{aligned}
F^{-1}\left[\sum_{i=1}^{N} w_i F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{N} \frac{q_i}{n^l} F(tx_i)\right] && \text{(plugging in } w_i = \tfrac{q_i}{n^l}\text{)} \\
&= F^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{N} q_i F(tx_i)\right] && \text{(rearranging)} \\
&= F^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{N}\sum_{j=1}^{q_i} F(tx_i)\right] && (q_i \in \mathbb{N}) \\
&= F^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{Q} F(t\tilde{x}_i)\right] && \text{(Definition of } \tilde{x}_i\text{)} \\
&= tF^{-1}\left[\frac{1}{n^l}\sum_{i=1}^{Q} F(\tilde{x}_i)\right] && \text{(Eq. (4.6))} \\
&= tF^{-1}\left[\sum_{i=1}^{N} w_i F(x_i)\right] && \text{(reversing the steps)}
\end{aligned}$$

Thus, we have derived

$$F^{-1}\left[\sum_{i=1}^{N} w_i F(tx_i)\right] = tF^{-1}\left[\sum_{i=1}^{N} w_i F(x_i)\right] \quad w_i \in \mathbb{W}. \tag{4.7}$$

Now, in order to approximate real weights, we need density of $\mathbb{W}$ in $\mathbb{R}_{\geq 0}$. Since n^l gets arbitrarily small, there exists for every $\varepsilon > 0$ an l such that $n^l < \varepsilon$. Therefore, if $x - y < \varepsilon$, for $x, y \in \mathbb{R}_{\geq 0}$ and $x > y$ there exists $k \in \mathbb{N}$ such that $x < \frac{k}{n^l} < y$. Thus, $\mathbb{W}$ is dense in $\mathbb{R}_{\geq 0}$ and we can approximate real weights with converging sequences from $\mathbb{W}$.

Finally, let $w_i \in \mathbb{R}_{\geq 0}$ and $q_{i_k} \in \mathbb{W}$ be sequences for $i = 1, \ldots, N$ such that $q_{i_k} \to w_i$ for $k \to \infty$. Then

$$\begin{aligned}
F^{-1}\left[\sum_{i=1}^{N} w_i F(tx_i)\right] &= F^{-1}\left[\sum_{i=1}^{N} \lim_{k\to\infty} q_{i_k} F(tx_i)\right] && \text{(Convergence of } q_{i_k}\text{)} \\
&= \lim_{k\to\infty} F^{-1}\left[\sum_{i=1}^{N} q_{i_k} F(tx_i)\right] && (F^{-1} \text{ continuous [5]}) \\
&= \lim_{k\to\infty} tF^{-1}\left[\sum_{i=1}^{N} q_{i_k} F(x_i)\right] && \text{(Eq. (4.7))} \\
&= tF^{-1}\left[\sum_{i=1}^{N} w_i F(x_i)\right]. && \text{(reversing the steps)}
\end{aligned}$$

Hence, the homogeneity holds for all $x_i, w_i, t \in \mathbb{R}_{\geq 0}$ and $N \in \mathbb{N}$, in particular, it holds for weights with sum 1. By Theorem 4.2.19 the only monotonic solutions are the power functions $\alpha x^p + \beta$ or the logarithm $\alpha \log(x)$ with $\alpha, \beta, p \in \mathbb{R}$ and $p > 0$. However, since $F(x) \geq 0$, it cannot be the logarithm and by $F(0) = 0$, the vertical shift β must be equal to 0. Therefore the only solution is, that the extended function takes the form αx^p, since we applied this theorem on the extension of F. Therefore $F(x) = \alpha x^p$ must hold at least on the interval $[0, \Omega)$.

Part 3 (Proving equality for positive reals)**.**
Now, for any $|\varphi_i(a_i) - \varphi_i(b_i)|$, we can find $t > 0$ small enough such that $t|\varphi_i(a_i) - \varphi_i(b_i)| < \omega$ for $i = 1, \ldots, n$. Applying monotonicity yields (we can assume $t \leq 1$):

$$\begin{aligned}
\delta(a, b) &= F^{-1}\left(\sum_{i=1}^{n} F\left(|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t} F^{-1}\left(\sum_{i=1}^{n} F\left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)\right) \\
&= \frac{1}{t}\left(\sum_{i=1}^{n} \left(t|\varphi_i(a_i) - \varphi_i(b_i)|\right)^p\right)^{1/p} && (\text{ since } t|\varphi_i(a_i) - \varphi_i(b_i)| \leq \omega) \\
&= \left(\sum_{i=1}^{n} |\varphi_i(a_i) - \varphi_i(b_i)|^p\right)^{1/p} \\
&= \delta_p\left(\varphi_i(a_i),\ \varphi_i(b_i)\right)
\end{aligned}$$

Thus, δ is a Minkowski metric and, by Theorem 2.2.7, the F_i are interval scales, allowing for affine linear transformations. However, since we set $F_i(0) = 0$, the affinity vanishes, and we are left with $F_i(x) = \alpha \cdot x^p$. Finally, the restriction $1 \leq p < \infty$ follows from the triangle inequality or, more precisely, from Minkowski's inequality; see (Bullen, 2003, pp. 189-191 Theorem 9).

$\square$

4.2.1 Homogeneity Theorem

Before beginning this section, we will first introduce some notation.
δ_∞ is used to represent the ∞-metric (also called the maximum metric) which is induced by the maximum norm[6]. This metric induces the natural topology of the euclidean space. $\mathrm{B}^{\delta}_{(r)} := \{x \in \mathbb{R}^n : \delta(x) < r\}$ denotes the open Ball with radius r around 0 with respect to the metric δ; similarly $\mathrm{B}^{\delta}_{[r]}$ denotes the closed ball. We use ∞ as a subscript, to indicate that the ball is with respect to δ_∞.

In this section we will prove the homogeneity theorem, which is needed for the proof of the main result, Theorem 4.2.1. The proof is divided into 5 parts and a whole subsection which is devoted to showing the convexity of δ-balls which is needed for the proof. First, we will state the theorem, give some remarks on it and prove some lemmas. Then we will prove the homogeneity theorem. Everything in this section is assumed as in this theorem. For example, if we write δ, this implies not any metric but the metric from this theorem with its properties. Moreover all topological terms(open, closed, continuous) are meant with respect to the natural topology of $\mathbb{R}^n$ (i.e. the topology induced by the δ_∞ and the euclidean metric), not the topology induced by δ or the induced topology on S. This will be especially important for the subsection on convexity of the balls.

It is suggested to first read this section (including the proof) to get the main ideas and a big picture understanding, and then read the next section.

For readers unfamiliar with convexity, we restate the definition of a convex set. The idea behind convexity is, that for every two points from a set, the line between these two points is also inside that set.

Definition 4.2.2 (Convexity)**.**
A set $A \subset \mathbb{R}^n$ is *convex*, iff $(1-\lambda)\mathbf{x} + \lambda\mathbf{y} \in A$ for all $\mathbf{x}, \mathbf{y} \in A$ and $\lambda \in [0,1]$.

First, we begin with the statement of the theorem.

Theorem 4.2.3 (Homogeneity Theorem)**.**
Let S be an open convex subset of n-dimensional Euclidean space. Let δ be a metric on S that satisfies intradimensional subtractivity (Eq. (2.2)) [7] *with respect to a continuous function $F : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ and linear coordinates* [8] *in S. If $\langle S, \delta\rangle$ is a*

[6]The maximum norm is defined by $\|x, y\|_\infty := \max_{i\leq n}\{|x_i - y_i|\}$.
[7]i.e. δ takes the form $\delta(\mathbf{x},\ \mathbf{y}) = F(|x_1 - y_1|, \ldots, |x_n - y_n|)$.
[8]Linear coordinates means that the vector representations are cartesian as opposed to, for example, polar, where each coordinate entry is not the value on the corresponding factor of the cartesian product.

metric space with additive segments, then δ *is homogeneous, i.e., for any* $\mathbf{x}, \mathbf{z} \in S$ *and any real* $0 \leq t \leq 1$*,*

$$\delta[\mathbf{x}, \mathbf{x} + t(\mathbf{z} - \mathbf{x})] = t\,\delta(\mathbf{x}, \mathbf{z}).$$

Now, we continue with some remarks that make it possible to move the set such that $\mathbf{0} \in S$ by using the translation invariance of δ.

Remark 4.2.4 (Translation invariance and domain of δ and F)*.* This remark establishes the continuity of δ and shows that we can assume without loss of generality, that $\mathbf{0} \in S$.

(i) δ is translation-invariant, i.e. $\delta\left(\mathbf{x} + \mathbf{y},\ \mathbf{z} + \mathbf{y}\right) = \delta\left(\mathbf{x},\ \mathbf{z}\right)$:

$$\begin{aligned}\delta\left(\mathbf{x} + \mathbf{y},\ \mathbf{z} + \mathbf{y}\right) &= F[|x_1 + y_1 - (z_1 + y_1)|, \ldots, x_n + |y_n - (z_n + y_n)|] \\ &= F[|x_1 - z_1|, \ldots, |x_n - z_n|] \\ &= \delta\left(\mathbf{x},\ \mathbf{z}\right)\end{aligned}$$

(ii) Because δ is invariant to translation in its domain, we can move S and redefine the metric on the shifted set. We do this by defining $S' = \{\mathbf{s} - \mathbf{x} : \mathbf{s} \in S\}$ for $\mathbf{x} \in S$ and $\delta'(\mathbf{a}, \mathbf{b}) := \delta\left(\mathbf{a} + \mathbf{x},\ \mathbf{b} + \mathbf{x}\right)$ for $\mathbf{a}, \mathbf{b} \in S'$. Since $\mathbf{0} \in S'$, we also can introduce the following shorthand Notation $\delta'(\mathbf{x}') := \delta'(\mathbf{0}, \mathbf{x}')$ for $\mathbf{x}' \in S'$. From now on, we assume that $\mathbf{0} \in S$, because by the argument just provided, we can shift S and δ such that $\mathbf{0} \in S$. Thus we can proceed to proof the statement using the shorthand notation, by which we need to show that:

$$\delta\left(t \cdot \mathbf{z}\right) = t \cdot \delta\left(\mathbf{z}\right) \qquad \forall t \in [0, 1] \text{ and } \forall \mathbf{z} \in S'.$$

(iii) Since δ is defined by $\delta\left(\mathbf{x},\ \mathbf{y}\right) = F(|x_1 - y_1|, \ldots, |x_n - y_n|)$ and δ is a metric, $F(\mathbf{x}) = 0$ iff $\mathbf{x} = 0$ must hold. Moreover the function F has to be defined on a set, where at least the "greatest possible difference" in coordinate i is a possible i-th argument. We denote this value as ρ_i and define it as

$$\rho_i := \sup\left\{|x_i - y_i| : (x_1, \ldots, x_n), (y_1, \ldots, y_n) \in S\right\} \in \mathbb{R}_{>0} \cup \{\infty\}.$$

Since δ is segmentally additive, every value between 0 and ρ_i must be in the domain of δ. Since this is true for all i, this implies that F is defined on a cartesian product of half-open, real intervals of the form $[0, \rho_i)$.

(iv) Moreover, due to the continuity of F and $|\cdot|$, the metric δ is continuous.

Before proceeding to the proof, we first establish two preliminary statements that will be needed at the beginning. The first one asserts that the distance between a compact and a closed set is greater than zero.

Lemma 4.2.5.
If $K \subset \mathbb{R}^n$ is compact, $C \subset \mathbb{R}^n$ is closed and K and C are disjoint, then there exists $\varepsilon > 0$ such that $\delta_\infty(\mathbf{p},\ \mathbf{y}) > \varepsilon$ for all $\mathbf{p} \in K$ and $\mathbf{p}' \in C$.

Proof. We prove this statement by contradiction. Assume that for all $\varepsilon > 0$ there exists $\mathbf{p} \in K$ and $\mathbf{p}' \in C$ such that $\delta_\infty(\mathbf{p},\ \mathbf{p}') < \varepsilon$. This means, we can construct two sequences $\mathbf{p_n} \in K$ and $\mathbf{p'_n} \in C$ such that $\delta_\infty(\mathbf{p_n},\ \mathbf{p'_n}) \to 0$ as $n \to \infty$. K is compact, thus there exists a convergent subsequence $\mathbf{p_{n_m}}$ such that $\mathbf{p_{n_m}} \to \mathbf{p} \in K$ for $n \to \infty$. With the triangle inequality we get

$$\delta_\infty\left(\mathbf{p},\ \mathbf{p'_{n_m}}\right) \leq \delta_\infty(\mathbf{p},\ \mathbf{p_{n_m}}) + \delta_\infty\left(\mathbf{p_{n_m}},\ \mathbf{p'_{n_m}}\right) \to \mathbf{0} \text{ as } m \to \infty.$$

Hence, $\mathbf{p}$ is a limit point of C, and since C is closed, $\mathbf{p} \in C$ must hold which contradicts the assumption. □

The second preliminary lemma allows us to make arbitrary small δ-balls which we will combine with the statement before to assure a distance in δ and not only in δ_∞.

Lemma 4.2.6 (arbitrary small δ-Balls)**.**
Let $\varepsilon > 0$ such that $\mathrm{B}^\infty_{(\varepsilon)} \subset S$. Then there exists $r > 0$ such that $\mathrm{B}^\delta_{(r)} \subset \mathrm{B}^\infty_{(\varepsilon)}$.

Proof. Since δ is a metric defined by $\delta(\mathbf{x},\mathbf{y}) = F[|x_1 - y_1|, \ldots, |x_n - y_n|]$, we can define the j-th partial function of F by setting all other arguments to zero, i.e. $F^j(x_j) := F(0, \ldots, x_j, \ldots, 0)$. F is strictly increasing in every argument implies that F^j is increasing and the continuity of F directly translates to F^j being continuous. Moreover, due to Remark 4.2.4 (iii), $F^j(x_j) = 0$ if and only if $x_j = 0$ holds. Together, this means that F^j is an increasing, continuous real function with $F^j(0) = 0$.

By setting $r_j = F^j(\varepsilon)$ we get by monotonicity, that $F^j(x_j) < r_j \implies x_j < \varepsilon$. Note, that $F^j(\varepsilon)$ is defined, because $\mathrm{B}^\infty_{(\varepsilon)} \subset S'$ implies that ε is a possible distance for each coordinate. (See also Remark 4.2.4 (iii). Using the a_i from the remark, we get that $\varepsilon < a_i$ holds for all i and thus $F^j(\varepsilon)$ is defined for all $j = 1, \ldots n$.) Doing this for $j = 1, \ldots, n$ yields $r_1, \ldots, r_n$. By setting $r = \min\{r_1, \ldots, r_n\}$ the above holds for every partial function F^j which yields the boundary for the ∞ Metric:

$$F^j(x_j) < r \leq r_j \implies x_j < \varepsilon \quad \forall j = 1, \ldots, n \implies \|\mathbf{x}\|_\infty < \varepsilon.$$

Using the monotonicity of F we get

$$\delta(\mathbf{x}) = F(\mathbf{x}) < r \implies F^j(x_j) < r \implies \|\mathbf{x}\|_\infty < \varepsilon.$$

$\square$

We now proceed to the proof of the homogeneity theorem. First, we restate the theorem in the form in which we will prove it.

Lemma (Restatement of the Homogeneity Theorem)**.**
Let S be an open convex subset of n-dimensional Euclidean space with $\mathbf{0} \in S$. Let δ be a metric on S that satisfies intradimensional subtractivity (Eq. (2.2)) with with respect to a continuous function $F : \mathbb{R}^n_{\geq 0} \to \mathbb{R}_{\geq 0}$ and linear coordinates in S. If $\langle S, \delta \rangle$ is a metric space with additive segments, then δ is homogeneous, i.e., for any $\mathbf{z} \in S$ and any real $0 \leq t \leq 1$,

$$\delta(t\mathbf{z}) = t\,\delta(\mathbf{z}).$$

The proof consists of 4 parts. First we will find a radius such that all balls with this radius on line from $\mathbf{0}$ to $\mathbf{z}$ are in S. Then we will prove the theorem for $t = \frac{1}{m}$ for $m \in \mathbb{N}$ by proving the two inequalities. After this, we will expand this statement to the rationals and then use continuity to expand it to all real numbers between 0 and 1.

Proof.

Part 0 (Asserting a common distance for all points on the line segment)**.**
Let $\mathbf{z} \in S$. Since S is open, $\mathbb{R}^n \setminus S$ is closed. The set $K := \{t\mathbf{z} : t \in [0,1]\}$ is compact, since continuous images($t \mapsto t\mathbf{z}$) of compact sets ($t \in [0,1]$) are compact. Thus, by Lemma 4.2.5 there exists a distance $\varepsilon > 0$ such that $\delta_\infty(\mathbf{p},\ \mathbf{p}') < \varepsilon$ for all $\mathbf{p} \in K$ and $\mathbf{p}' \in \mathbb{R}^n \setminus S$. This means, for all points on the line from $\mathbf{0}$ to $\mathbf{z}$ the ball with radius ε is a subset of S, i.e. for all $t \in [0,1] : \mathrm{B}^\infty_\varepsilon(t\mathbf{z}) \subset S$. By Lemma 4.2.6, there exists $r > 0$ such that $\mathrm{B}^\delta_{(r)} \subset \mathrm{B}^\infty_{(\varepsilon)}$, and by translation invariance this holds for all points on the line: $\mathrm{B}^\delta_r(t\mathbf{z}) \subset S$ for all $t \in [0,1]$. Now let $m \in \mathbb{N}$ such that $\frac{1}{m}\,\delta(\mathbf{z}) < r$. In particular, this means that $\mathrm{B}^\delta_{(r)}$ and therefore, $\mathrm{B}^\delta_{(\frac{1}{m}\,\delta(\mathbf{z}))}$ is bounded (with respect to the ∞ metric).

Part 1 ($t\,\delta(\mathbf{z}) \leq \delta(t\mathbf{z})$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$)**.**
Let $\mathbf{z^{(k)}} := \frac{k}{m} \cdot \mathbf{z}$ for $k = 0, \ldots, m$. Then $\mathbf{z^{(k)}} \in S$ for all k due to convexity of

S. By applying translation invariance, we can calculate the distance between two consecutive points:

$$\delta\left(\mathbf{z^{(k-1)}},\ \mathbf{z^{(k)}}\right) = \delta\left(\frac{k-1}{m}\mathbf{z},\ \frac{k-1}{m}\mathbf{z} + \frac{1}{m}\mathbf{z}\right) = \delta\left(\frac{1}{m}\mathbf{z}\right) \quad \forall k = 1, \ldots, m.$$

Applying the triangle inequality $m-1$-times and using the above result yields

$$\delta\left(\mathbf{z}\right) \leq \sum_{k=0}^{m-1} \delta\left(\mathbf{z^{(k)}},\ \mathbf{z^{(k+1)}}\right) = m \cdot \delta\left(\frac{1}{m}\mathbf{z}\right).$$

Finally, dividing the right side by m yields $\frac{1}{m}\,\delta\left(\mathbf{z}\right) \leq \delta\left(\frac{1}{m}\mathbf{z}\right)$, which concludes the first part.

Part 2 ($t\,\delta\left(\mathbf{z}\right) \geq \delta\left(t\mathbf{z}\right)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$)**.**
Since δ is a metric with additive segments, there exists a segment from $\mathbf{0}$ to $\mathbf{z}$ with length $\delta\left(\mathbf{z}\right)$. By Remark 3.1.2 the arc length is a continuous, non-decreasing function. This means, we can divide the segment into m subsegments with length $\frac{1}{m}\,\delta\left(\mathbf{z}\right)$ and choose $\mathbf{y^{(0)}}, \ldots, \mathbf{y^{(m)}}$ equally spaced on the segment such that $\delta\left(\mathbf{y^{(k)}},\ \mathbf{y^{(k-1)}}\right) = \frac{1}{m}\,\delta\left(\mathbf{z}\right)$ and $\mathbf{y^{(0)}} = \mathbf{0}$ and $\mathbf{y^{(m)}} = \mathbf{z}$. Using this, we can define $\mathbf{x^{(k)}} := \mathbf{y^{(k)}} - \mathbf{y^{(k-1)}}$. Translation invariance yields $\delta\left(\mathbf{x^{(k)}}\right) = \delta\left(\mathbf{y^{(k)}},\ \mathbf{y^{(k-1)}}\right) = \frac{1}{m}\,\delta\left(\mathbf{z}\right)$ and, due to the choice of m in Part 0, $\mathbf{x^{(k)}} \in S$ holds. Note, that this is still true, if we increase m.

Further, we can observe that $\frac{1}{m}\mathbf{z}$ is a convex combination of $\mathbf{x^{(k)}}$:

$$\sum_{k=1}^{m} \frac{1}{m}\mathbf{x^{(k)}} = \frac{1}{m}(\mathbf{y^{(1)}} - \mathbf{y^{(0)}} + \mathbf{y^{(2)}} - \mathbf{y^{(1)}} + \cdots + \mathbf{y^{(m)}}) = \frac{1}{m}(-\mathbf{y^{(0)}} + \mathbf{y^{(m)}}) = \frac{1}{m}\mathbf{z}$$

We showed before, that $\delta\left(\mathbf{x^{(k)}}\right) = \frac{1}{m}\,\delta\left(\mathbf{z}\right)$, which implies $\mathbf{x^{(k)}} \in \mathrm{B}^{\delta}_{[\frac{1}{m}\,\delta(\mathbf{z})]}$. By Part 0 $\mathrm{B}^{\delta}_{(\frac{1}{m}\,\delta(\mathbf{z}))}$ is bounded. Therefore we can apply Lemma 4.2.7 from which we get that spheres are convex. This means, that the spheres are closed under convex combination, which implies that $\frac{1}{m}\mathbf{z} \in \mathrm{B}^{\delta}_{[\frac{1}{m}\,\delta(\mathbf{z})]}$ must hold. Thus $\delta\left(\frac{1}{m}\mathbf{z}\right) \leq \frac{1}{m}\,\delta\left(\mathbf{z}\right)$ which concludes the second part of the proof.

Now, by combining Part 1 and Part 2, we have established the equality for $t = \frac{1}{m}$. Note, that the equality also holds for all $\mathbb{N} \ni m' > m$ since we only needed m to be big enough in Part 2 for $\mathbf{x^{(k)}}$ to be in S and for convexity of the spheres. Thus, by making m bigger, the crucial points in Part 2 and therefore the equality are still true. Next, we will proceed to extending the equality from $t = \frac{1}{m}$ to $t = \frac{k}{m}$ for $k = 0, \ldots, m$.

Part 3 ($t\,\delta(\mathbf{z}) = \delta(t\mathbf{z})$ for $t = \frac{k}{m}$ and $k \in \{0, \ldots, m\}$)**.**
For $\mathbf{z^{(1)}}$ we already have $\delta\left(\mathbf{0},\ \mathbf{z^{(1)}}\right) = \frac{1}{m}\,\delta(\mathbf{0},\ \mathbf{z})$. Now, for consecutive points, by translation invariance, it holds that

$$\delta\left(\mathbf{z^{(k-1)}},\ \mathbf{z^{(k)}}\right) = \delta\left(\mathbf{z^{(k-2)}},\ \mathbf{z^{(k-1)}}\right) = \cdots = \delta\left(\mathbf{z^{(1)}},\ \mathbf{z^{(0)}}\right) = \delta\left(\mathbf{z^{(1)}},\ \mathbf{0}\right) = \frac{1}{m}\,\delta(\mathbf{0},\ \mathbf{z}).$$

Using this and the triangle inequality, we get

$$\begin{aligned} \delta(\mathbf{0},\ \mathbf{z}) &\le \delta\left(\mathbf{0},\ \mathbf{z^{(1)}}\right) + \delta\left(\mathbf{z^{(1)}},\ \mathbf{z^{(2)}}\right) + \cdots + \delta\left(\mathbf{z^{(m-1)}},\ \mathbf{z^{(m)}}\right) \\ &= n \cdot \frac{1}{n}\,\delta(\mathbf{0},\ \mathbf{z}) \\ &= \delta(\mathbf{0},\ \mathbf{z}). \end{aligned}$$

This implies

$$\delta(\mathbf{0},\ \mathbf{z}) = \delta\left(\mathbf{0},\ \mathbf{z^{(1)}}\right) + \delta\left(\mathbf{z^{(1)}},\ \mathbf{z^{(2)}}\right) + \cdots + \delta\left(\mathbf{z^{(m-1)}},\ \mathbf{z^{(m)}}\right). \tag{4.8}$$

For the distance $\delta\left(\mathbf{0},\ \mathbf{z^{(2)}}\right)$, we get the following two inequalities.
Firstly,

$$\delta\left(\mathbf{0},\ \mathbf{z^{(2)}}\right) \le \delta\left(\mathbf{0},\ \mathbf{z^{(1)}}\right) + \delta\left(\mathbf{z^{(1)}},\ \mathbf{z^{(2)}}\right) = \frac{2}{m}\,\delta(\mathbf{0},\ \mathbf{z}).$$

Secondly,

$$\begin{aligned} \delta(\mathbf{0},\ \mathbf{z}) &\le \delta\left(\mathbf{0},\ \mathbf{z^{(2)}}\right) + \delta\left(\mathbf{z^{(2)}},\ \mathbf{z}\right) \\ \delta\left(\mathbf{0},\ \mathbf{z^{(2)}}\right) &\ge \delta(\mathbf{0},\ \mathbf{z}) - \underbrace{\delta\left(\mathbf{z^{(2)}},\ \mathbf{z}\right)}_{\le \sum_k \delta\left(\mathbf{z^{(2+k)}},\ \mathbf{z^{(2+k+1)}}\right)} && \text{(Rearrange)} \\ &\ge \delta(\mathbf{0},\ \mathbf{z}) - \sum_{k=0}^{m-3} \delta\left(\mathbf{z^{(2+k)}},\ \mathbf{z^{(2+k+1)}}\right) && (\triangle\ \text{inequality}) \\ &= \delta(\mathbf{0},\ \mathbf{z}) - \frac{m-2}{m}\,\delta(\mathbf{0},\ \mathbf{z}) && \text{(using Eq. (4.8))} \\ &= \frac{2}{m}\,\delta(\mathbf{0},\ \mathbf{z}). \end{aligned}$$

Combining these two inequalities yields $\delta\left(\mathbf{0},\ \mathbf{z^{(2)}}\right) = \frac{2}{m}\,\delta(\mathbf{0},\ \mathbf{z})$. By repeating this for $\delta\left(\mathbf{0},\ \mathbf{z^{(k)}}\right)$ for every $k = 2, \ldots, m-1$, we get the equality for every rational number for m big enough. However every rational number with a denominator smaller than m can be represented by another rational number with denominator greater than m. This yields, that the result is true for every rational t.

Part 4 ($t\,\delta(\mathbf{z}) = \delta(t\mathbf{z})$ for $t \in [0, 1]$)**.**
The final step can be completed using continuity of the metric. Now let $t \in [0, 1]$

and $t_n \in \mathbb{Q}$ with $t_n \to t$ for $n \to \infty$. Then

$$\begin{aligned}
\delta\left(t\mathbf{z}\right) &= \delta\left(\lim_{n\to\infty} t_n\mathbf{z}\right) \\
&= \lim_{n\to\infty} \delta\left(t_n\mathbf{z}\right) && \text{(continuity)} \\
&= \lim_{n\to\infty} t_n\,\delta\left(\mathbf{z}\right) && (t_n \in \mathbb{Q} \text{ and Part 3}) \\
&= t\,\delta\left(\mathbf{z}\right)
\end{aligned}$$

□

Convexity of δ Balls

In this subsection we will proof the following lemma which is needed to conclude the proof of Theorem 4.2.3. Everything will be assumed as in Theorem 4.2.3. Note, that all topological terms are with respect to the natural topology of the euclidean space.

Lemma 4.2.7 (Convexity of δ-Spheres)**.**
For $r > 0$ if $\mathrm{B}^{\delta}_{[r]} \subset S$ is bounded, then it is convex.

First, we will state some definitions and results on basic convex geometry that we need, and prove the lemma at the end of this section. The definitions and results on convex geometry are taken out of (Hug & Weil, 2020).

First of all, we need the notions of convex combinations, the convex hull and extreme points. Recall, that a set is convex if for all two pairs of points of that set, the linear interpolation between these points stays inside the set. By increasing the amount of points, we get what is called the convex combination of these points. A convex combination is similar to a linear combination, but has the additional constraint that the coefficients must be positive and add up to 1. The convex hull of a set is, like any other hull, the smallest convex set that contains the set.

Definition 4.2.8 (Convex Combination and Hull)**.**

1. Let $k \in \mathbb{N}$, let $\mathbf{x_1}, \dots, \mathbf{x_k} \in \mathbb{R}^n$, and let $\lambda_1, \dots, \lambda_k \in [0, 1]$ be such that $\lambda_1 + \cdots + \lambda_k = 1$. Then $\lambda_1\mathbf{x_1} + \cdots + \lambda_k\mathbf{x_k}$ is called a *convex combination* of the points $\mathbf{x_1}, \dots, \mathbf{x_k}$.

2. For $A \subset \mathbb{R}^n$ the *convex Hull* $\mathrm{conv}(A)$ is the smallest convex set containing A, i.e.
$$\mathrm{conv}(A) := \cap \{C \subset \mathbb{R}^n : A \subset C, C \text{ convex}\}$$

The next theorem sums up the relationship between convex combination and convex hulls.

Theorem 4.2.9.
For $A \subset \mathbb{R}^n$ the convex Hull is the set of all convex combination of points in A:

$$\mathrm{conv}(A) = \left\{ \sum_{i=1}^{k} \lambda_i \mathbf{x_i} : k \in \mathbb{N}, \mathbf{x_1}, \dots, \mathbf{x_k} \in A, \lambda_1, \dots, \lambda_k \in [0, 1] : \sum_{i=1}^{k} \lambda_i = 1 \right\}$$

Moreover, the convex hull preserves compactness, as the following theorem shows.

Theorem 4.2.10 (convex Hull of compact set is compact).
If $A \subset \mathbb{R}^n$ is compact, then $\operatorname{conv}(A)$ *is compact.*

Next, we will introduce extreme points, which are, in a sense, characteristic points of a set that cannot be represented by a (strict) convex combination.

Definition 4.2.11 (Extreme Point).
Let $A \subset \mathbb{R}^n$ be closed and convex. A point $\mathbf{x} \in A$ is called an *extreme point of A* if $\mathbf{x}$ cannot be represented as a nontrivial (or strict) convex combination of points of A, that is, if $\mathbf{x} = \lambda \cdot \mathbf{y} + (1-\lambda) \cdot \mathbf{z}$ with $\mathbf{y}, \mathbf{z} \in A$ and $\lambda \in (0,1)$, implies that $\mathbf{x} = \mathbf{y} = \mathbf{z}$. The set of all extreme points of A is denoted by $\operatorname{extreme}(A)$.

For compact and convex sets, the extreme points are enough to "build" the convex hull of the set by doing convex combinations. In particular, this means that for a compact and convex set, every point can be expressed as a convex combination of extreme points.

Theorem 4.2.12 (Minkowski Theorem).
Let $A \subset \mathbb{R}^n$ be compact and convex. Then $A = \operatorname{conv}(\operatorname{extreme} A)$.

Before proving the convexity of δ balls, we will first establish two lemmas that will be needed in the proof. The first one is the openness/closedness of δ-balls.

Lemma 4.2.13 (δ-Balls are open/closed).
For every radius $r > 0$, the δ-Balls are open $(\mathrm{B}^{\delta}_{(r)})$ *or closed* $(\mathrm{B}^{\delta}_{[r]})$.

Proof. $\delta : S' \times S' \to \mathbb{R}_{\geq 0}$ is continuous with $S' \subseteq \mathbb{R}^n$ and therefore $\mathbf{y} \mapsto \delta(\mathbf{y}) = \delta(\mathbf{0},\ \mathbf{y})$ is continuous. Consider the closed interval $U' = [0, r] \subset \mathbb{R}_{\geq 0}$. Then

$$\begin{aligned}
\delta^{-1}(U') &= \{\mathbf{y} \in S' : \delta(\mathbf{y}) \in U'\} && \text{(Preimage definition)}\\
&= \{\mathbf{y} \in S' : \delta(\mathbf{0},\ \mathbf{y}) \leq r\} && \text{(Definition of } \delta)\\
&= \mathrm{B}^{\delta}_{[r]}\,.
\end{aligned}$$

Due to topological continuity of δ (which is equivalent to preimages of closed sets being closed) $\mathrm{B}^{\delta}_{[r]}$ is closed. By translation invariance this holds for closed balls with any center in S. Moreover, we can obtain the same result for open balls by repeating this proof and replacing the closed interval with an open one and $\leq$ with $<$. $\square$

The second lemma shows, that extreme points of balls get compressed by a factor of $\frac{1}{m} \in \mathbb{N}$, if the radius of the ball gets shrinked by the same factor.

Lemma 4.2.14 (compressing extreme points)**.**
Let $r \in \mathbb{R}_{>0}$ and $m \in \mathbb{N}_{>0}$. Further let $\mathrm{E}\,(r) := \mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})$ and $\mathrm{C}\,[r] := \mathrm{conv}(\mathrm{B}^{\delta}_{[r]})$. Then, $\mathbf{y} \in \mathrm{E}\,(r)$ implies $\frac{1}{m}\mathbf{y} \in \mathrm{E}\left(\frac{r}{m}\right)$.

Remark. We use square brackets for $\mathrm{C}\,[r]$ to denote that this set is closed. For the reasoning, see the first argument in Section 4.2.1.

Proof. We prove this statement by contraposition.

Assume, that $\frac{1}{m}\mathbf{y}$ is not an extreme point, i.e. $\frac{1}{m}\mathbf{y} \in \mathrm{C}\left[\frac{r}{m}\right] \setminus \mathrm{E}\left(\frac{r}{m}\right)$. Then we can write $\frac{1}{m}\mathbf{y}$ as a strict convex combination of elements $\mathbf{x^{(k)}} \in \mathrm{C}\left[\frac{r}{m}\right]$:

$$\frac{1}{m}\mathbf{y} = \sum_{i=1}^{p} \lambda_i \mathbf{x^{(k)}} \quad \text{with} \sum_{i=1}^{p} \lambda_i = 1, \quad \lambda_i > 0,\ \mathbf{x^{(k)}} \in \mathrm{C}\left[\frac{r}{m}\right] \text{ and } p \in \mathbb{N}$$

But because $\mathrm{C}\left[\frac{r}{m}\right] = \mathrm{conv}(\mathrm{B}^{\delta}_{[\frac{r}{m}]})$ holds, we can write any element from $\mathrm{C}\left[\frac{r}{m}\right]$ as a convex combination of elements from $\mathrm{B}^{\delta}_{[\frac{r}{m}]}$ and therefore we can assume $\mathbf{x^{(k)}} \in \mathrm{B}^{\delta}_{[\frac{r}{m}]}$. By translation invariance and the triangle inequality we get for all i:

$$\begin{aligned}
\delta\left(m\mathbf{x^{(k)}}\right) &\leq \delta\left(0,\ \mathbf{x^{(k)}}\right) + \delta\left(\mathbf{x^{(k)}},\ m\mathbf{x^{(k)}}\right) && \text{triangle inequality}\\
&= \delta\left(\mathbf{x^{(k)}}\right) + \delta\left(0,\ (m-1)\cdot\mathbf{x^{(k)}}\right) && \text{translation invariance}\\
&\leq \delta\left(\mathbf{x^{(k)}}\right) + \delta\left(0,\ \mathbf{x^{(k)}}\right) + \delta\left(\mathbf{x^{(k)}},\ (m-1)\cdot\mathbf{x^{(k)}}\right) && \text{triangle inequality}\\
&= 2\cdot\delta\left(\mathbf{x^{(k)}}\right) + \delta\left(0,\ (m-2)\cdot\mathbf{x^{(k)}}\right) && \text{translation invariance}\\
&\ldots\\
&\leq m\cdot\delta\left(\mathbf{x^{(k)}}\right)\\
&= m\cdot\frac{r}{m} = r\\
\Longrightarrow\ m\mathbf{x^{(k)}} &\in \mathrm{B}^{\delta}_{[r]}
\end{aligned}$$

This yields, that $\mathbf{y} = \sum_{i=1}^{p} \lambda_i m \mathbf{x^{(k)}}$ is a strict convex combination (because $\lambda_i m \neq 0$) and therefore $\mathbf{y}$ is not in $\mathrm{E}\,(r)$. □

Now we can prove the convexity of δ-balls.

Proof of Lemma 4.2.7. Let $\mathrm{E}\,(r) := \mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})$ and $\mathrm{C}\,[r] := \mathrm{conv}(\mathrm{B}^{\delta}_{[r]})$. By Lemma 4.2.13 $\mathrm{B}^{\delta}_{[r]}$ is closed and by assumption it is also bounded. Therefore, from the Heine-Borel Theorem, it follows that $\mathrm{B}^{\delta}_{[r]}$ is compact. By definition and Theorem 4.2.10 it follows, that $\mathrm{C}\,[r]$ is compact (and therefore closed, which is the reason why we denote it with square brackets) and convex. Application of the Minkowski Theorem (Theorem 4.2.12) yields

$$\mathrm{C}\,[r] = \mathrm{conv}(\mathrm{extreme}(\mathrm{C}\,[r])).$$

Further, it holds that $\mathrm{extreme}(\mathrm{C}\,[r]) = \mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})$ (By doing convex combinations, no new extreme points can be introduced because these are the points, which cannot be represented by a convex combination). Combining this and the statement above yields:

$$\mathrm{B}^{\delta}_{[r]} \subset \mathrm{C}\,[r] = \mathrm{conv}(\mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})).$$

Now, if we prove that $\mathrm{conv}(\mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})) \subset \mathrm{B}^{\delta}_{[r]}$, we get that $\mathrm{B}^{\delta}_{[r]} = \mathrm{C}\,[r]$ which proves the convexity of $\mathrm{B}^{\delta}_{[r]}$. We simplify this statement before proving it.

$$\mathrm{conv}(\underbrace{\mathrm{extreme}(\mathrm{B}^{\delta}_{[r]})}_{\mathrm{E}(r)}) \subset B$$

$$\Updownarrow$$

$$\sum_{i=1}^{k} \lambda_i \mathbf{y^{(i)}} \in \mathrm{B}^{\delta}_{[r]}, \quad k \in \mathbb{N}_{>0}, \begin{cases} \mathbf{y^{(i)}} \in \mathrm{E}\,(r) \\ \lambda_i \in \mathbb{R}_{\geq 0}, \sum_{i=1}^{k} \lambda_i = 1 \end{cases} \qquad \text{(Definition)}$$

$$\Uparrow$$

$$\sum_{i=1}^{k} \lambda_i \mathbf{y^{(i)}} \in \mathrm{B}^{\delta}_{[r]}, \quad k \in \mathbb{N}_{>0}, \begin{cases} \mathbf{y^{(i)}} \in \mathrm{E}\,(r) \\ \lambda_i \in \mathbb{Q}_{\geq 0}, \sum_{i=1}^{k} \lambda_i = 1 \end{cases} \qquad (B \text{ closed, } \mathbb{Q} \subset \mathbb{R} \text{ dense } ^9)$$

$$\Uparrow$$

$$\sum_{i=1}^{m} \frac{1}{m} \mathbf{y^{(i)}} \in \mathrm{B}^{\delta}_{[r]}, \quad m \in \mathbb{N}_{>0},\ \mathbf{y^{(i)}} \in \mathrm{E}\,(r) \qquad (m \text{ common denominator})$$

Thus, we will continue proving that for $m \in \mathbb{N}_{>0}$ and $\mathbf{y^{(i)}} \in \mathrm{E}\,(r)$ it holds that $\sum_{i=1}^{m} \frac{1}{m}\mathbf{y^{(i)}} \in \mathrm{B}^{\delta}_{[r]}$.

For $\mathbf{y^{(i)}} \in \mathrm{E}\,(r)$ it holds by Lemma 4.2.14 that $\frac{1}{m}\mathbf{y^{(i)}} \in \mathrm{E}\left(\frac{r}{m}\right) \subset \mathrm{B}^{\delta}_{[\frac{r}{m}]}$. This means, that for the metric distance of the convex combination the following holds:

$$
\begin{aligned}
\delta\left(\mathbf{0},\ \sum_{i=1}^{m}\frac{1}{m}\mathbf{y^{(i)}}\right) &\le \delta\left(\mathbf{0},\ \frac{1}{m}\mathbf{y^{(1)}}\right) + \delta\left(\frac{1}{m}\mathbf{y^{(1)}},\ \sum_{i=1}^{m}\frac{1}{m}\mathbf{y^{(i)}}\right) \\
&\qquad \text{(triangle inequality)} \\
&= \delta\left(\mathbf{0},\ \frac{1}{m}\mathbf{y^{(1)}}\right) + \delta\left(\mathbf{0},\ \sum_{i=2}^{m}\frac{1}{m}\mathbf{y^{(i)}}\right) \quad \text{(translation invariance)} \\
&\le \frac{r}{m} + \delta\left(\mathbf{0},\ \sum_{i=2}^{m}\frac{1}{m}\mathbf{y^{(i)}}\right) \quad \text{(translation invariance)} \\
&\dots \quad \text{(repeat)} \\
&\le m\frac{r}{m} = r
\end{aligned}
$$

This yields that $\sum_{i=1}^{m}\frac{1}{m}\mathbf{y^{(i)}} \in \mathrm{B}^{\delta}_{[r]}$ holds which proves the statement. □

4.2.2 Uniqueness Theorem for Homogeneous Means

In this section we will prove the uniqueness theorem for homogeneous means, which states that the power means [10] (including the geometric mean as a limit case) are the only homogeneous quasi-arithmetic means. Before beginning the proof, we will first define quasi-arithmetic means and one class of this means, the power means. Both, the proof and the definitions are based on (Hardy et al., 1952, pp. 66-69) and (Bullen, 2003, p. 271-273).

We begin with defining quasi-arithmetic means. The name stems from generalizing the arithmetic mean by applying a function on each value before averaging and then using its inverse to transform the result back.

Definition 4.2.15 (Quasi Arithmetic Means and Homogeneity)**.**
Let $\phi : [0, \infty) \to [0, \infty)$ be continuous and strictly increasing and let $\mathbf{x}, \mathbf{w} \in \mathbb{R}^n_{\geq 0}$. Then the quasi-arithmetic ϕ-mean of $\mathbf{x}$ with weights $\mathbf{w}$

$$\mathcal{M}_\phi(\mathbf{x},\ \mathbf{w}) = \phi^{-1}\left(\sum_{i=1}^{n} w_i \phi(x_i)\right)$$

The mean is said to be *homogeneous*, if

$$\mathcal{M}_\phi(t \cdot \mathbf{x},\ \mathbf{w}) = t \cdot \mathcal{M}_\phi(\mathbf{x},\ \mathbf{w})$$

holds for all $t > 0$.

Next, we define the power means and remark that they are quasi-arithmetic means.

Definition 4.2.16 (Power mean)**.**
For $n \in \mathbb{N}, \mathbf{x}, \mathbf{w} \in \mathbb{R}^n_{\geq 0}$ and $p > 0$ we define the power means $\mathcal{M}_p(\mathbf{x}, \mathbf{w})$ as

$$\mathcal{M}_p(\mathbf{x}, \mathbf{w}) = \left(\frac{1}{\sum_{i=1}^{n} w_i} \sum_{i=1}^{n} w_i x_i^p\right)^{1/p}.$$

For $p = 0$ we can define $\mathcal{M}_0(\mathbf{x}, \mathbf{w})$ as the geometric mean

$$\mathcal{M}_0(\mathbf{x}, \mathbf{w}) = \left(\prod_{i=1}^{n} x_i^{w_i}\right)^{1/\sum_{i=1}^{n} w_i}$$

since it is obtained as a limit case for $r \to 0$ (Bullen, 2003, p. 176).

[10] In some literature this is also referred to as the Hölder mean or the generalized means.

Remark 4.2.17 (Power Means are Quasi-Arithmetic). By setting $\phi(x) = x^p$ for $p > 0$ or $\phi(x) = \log(x)$ we obtain the power means for $p > 0$ and $p = 0$ respectively.
In particular, setting $w_i = 1$ yields the familiar generalizations of the power mean

$$\left(\frac{1}{n}\sum_{i=1}^{n} x_i^p\right)^{1/p}.$$

and the arithmetic mean

$$\sqrt[n]{\prod_{i=1}^{n} x_i}.$$

Now we can continue with a theorem about equal means, which is needed in the proof of the uniqueness theorem for homogeneous means.

Theorem 4.2.18 (Characterization of Equal Means)**.**
In order that

$$\mathcal{M}_\chi(\mathbf{a},\ \mathbf{w}) = \mathcal{M}_\psi(\mathbf{a},\ \mathbf{w}) \tag{4.9}$$

for all $\mathbf{a}, \mathbf{w}$ *and* $n \in \mathbb{N}$*, it is necessary and sufficient that*

$$\phi = \alpha \cdot \psi + \beta \tag{4.10}$$

where α *and* β *are constants and* $\alpha \neq 0$*.*

Proof.

Part 1 ($\Rightarrow$)**.**
Assume that Equation (4.9) holds. In particular, it also holds for $n = 2$. Thus we have

$$\chi^{-1}(w_1\chi(a_1) + w_2\chi(a_2)) = \psi^{-1}(w_1\psi(a_1) + w_2\psi(a_2))$$

Putting $\phi(x) = \chi(\psi^{-1}(x)), x_i = \psi(a_i)$ for $i = 1, 2$ yields

$$\phi(w_1x_1 + w_2x_2) = w_1\psi(x_1) + w_2\psi(x_2) \quad x_1, x_2 \in \mathbb{R}_{\geq 0}.$$

In particular, this holds for $w_1, w_2 = 1/2$ which then reduces to

$$\phi\left(\frac{x_1 + x_2}{2}\right) = \frac{\phi(x_1)}{2} + \frac{\phi(x_2)}{2} \quad x_1, x_2 \in \mathbb{R}_{\geq 0}.$$

By (Aczél, 1966, p. 46), the general continuous solution is $\psi(x) = \alpha x + \beta$, hence $\chi = \alpha\psi + \beta$.

Part 2 ($\Leftarrow$).
Assume that Equation (4.10) holds. Then

$$\begin{aligned}
\chi(\mathcal{M}_\chi(\mathbf{a},\ \mathbf{w})) &= \sum_{i=1}^{n} w_i \chi(a_i) && \text{(cancellation of inverses)} \\
&= \sum_{i=1}^{n} q_i \alpha \psi(a_i) + \beta && \text{(Eq. (4.10))} \\
&= \alpha \left(\sum_{i=1}^{n} w_i \psi(a_i) \right) + \beta && \text{(rearranging)} \\
&= \chi \left[\mathcal{M}_\psi(\mathbf{a},\ \mathbf{w})\right] && \text{(Eq. (4.10))}
\end{aligned}$$

Applying χ^{-1} to both sides of the equation yields Equation (4.9).

□

Now we finally state and prove the main theorem of this section.

Theorem 4.2.19 ($\mathcal{M}_p$ are the only homogeneous means $\mathcal{M}_\phi$)**.**
Suppose that ϕ is continuous and strictly increasing in the open interval $(0, \infty)$ and that the mean defined by ϕ is homogeneous, i.e.

$$\mathcal{M}_\phi(t\mathbf{a},\ \mathbf{w}) = t\mathcal{M}_\phi(\mathbf{a},\ \mathbf{w}) \tag{4.11}$$

for all positive $\mathbf{a}, \mathbf{w}, t$ and all $n \in \mathbb{N}$ such that $\sum_{i=1}^{n} w_i = 1$. Then $\mathcal{M}_\phi(\mathbf{a},\ \mathbf{w})$ is $\mathcal{M}_p(\mathbf{a}, \mathbf{w})$.

Proof. By Theorem 4.2.18 we can assume that $\phi(1) = 0$, since we can replace $\phi(x)$ by $\phi(x) - \phi(1)$ without changing $\mathcal{M}_\phi(\mathbf{a},\ \mathbf{w})$. Next, we write Equation (4.11) as

$$\mathcal{M}_\phi(\mathbf{a},\ \mathbf{w}) = t^{-1}\mathcal{M}_\phi(t\mathbf{a},\ \mathbf{w}) = t^{-1}\phi^{-1}\left(\sum_{i=1}^{n} w_i \phi(ta_i)\right) = \mathcal{M}_\psi(\mathbf{a},\ \mathbf{w})$$

by setting $\psi(x) = \phi(tx)$. From this, it follows that $\psi^{-1}(x) = t^{-1} \cdot \phi(x)$ [11], which in turn justifies the above equation. Therefore, we have the equality of the means defined by ψ and ϕ. Since ψ is defined for every $t > 0$, we have by Theorem 4.2.18 that there are $\alpha(t), \beta(t) \in \mathbb{R}, \alpha(t) \neq 0$ such that

$$\phi(tx) = \psi(x) = \alpha(t) \cdot \phi(x) + \beta(t).$$

[11] For this, let $\psi^{-1}(x) = y$. Then, by applying ψ to both sides, and using its definition, this yields $x = \phi(ty)$. Applying the inverse of ϕ and dividing by t yields $t^{-1}\phi^{-1}(x) = y$, which in turn is equal to $\psi^{-1}(x)$ by the first equation.

Plugging in $x = 1$ and using $\phi(1) = \phi(0)$ yields

$$\phi(t) = \psi(1) = \beta(t).$$

Thus, for all positive x, y by setting $y = t$

$$\phi(xy) = \alpha(y)\phi(x) + \phi(y) \tag{4.12}$$

Similarly, by exchanging the roles of x and y we get

$$\phi(xy) = \alpha(x)\phi(y) + \phi(x)$$

Combining these two equalities yields

$$\begin{aligned} \alpha(y)\phi(x) + \phi(y) &= \alpha(x)\phi(y) + \phi(x) && (-\phi(x) - \phi(y)) \\ \alpha(y)\phi(x) - \phi(x) &= \alpha(x)\phi(y) - \phi(y) && (\text{divide }^{12}\text{by } \phi(x) \cdot \phi(y)) \\ \frac{\alpha(x) - 1}{\phi(x)} &= \frac{\alpha(y) - 1}{\phi(y)} := c \end{aligned}$$

Since this must hold for all x and y, the expressions must reduce to a constant c. This can be seen by settings, for example, $x = 1$ and observing that the right hand side must equal to that for all y and therefore must be constant. The same argument can be applied in reverse. Using this, we get the following expression

$$\alpha(y) = 1 + c\phi(y).$$

Plugging this expression for $\alpha(y)$ into Equation (4.12) yields

$$\phi(xy) = c\phi(x)\phi(y) + \phi(x) + \phi(y). \tag{4.13}$$

For this equation, there are two cases to consider: $c = 0$ and $c \neq 0$. If $c = 0$, then Equation (4.13) reduces to the well known logarithmic Cauchy equation

$$\phi(xy) = \phi(x) + \phi(y),$$

of which the most general and for $x > 0$ continuous solution is $\phi(x) = \alpha \log(x)$ for some constant $\alpha \in \mathbb{R}$; see (Aczél, 1966, p. 39) and (Cauchy, 1821, p. 110-111).

If $c \neq 0$, we set $f(x) = c\phi(x)+1$ and Equation (4.13) reduces to the multiplicative Cauchy functional equation

$$f(xy) = f(x)f(y),$$

whose general solution is $f(x) = x^r$; see (Aczél, 1966, p. 39) and (Cauchy, 1821, p. 111-112). Substituting back yields $\phi(x) = \frac{x^r-1}{c}$. □

[12] Provided $x \neq 1, y \neq 1$. Since Equation (4.13) is true when x or y is 1, we can ignore the exception.

Sources and Original Contributions

sources and contribs

Chapter 5

Discussion

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Bei einer Thesis des Fachbereichs Architektur entspricht die eingereichte elektronische Fassung dem vorgestellten Modell und den vorgelegten Plänen.

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For a thesis of the Department of Architecture, the submitted electronic version corresponds to the presented model and the submitted architectural plans.

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